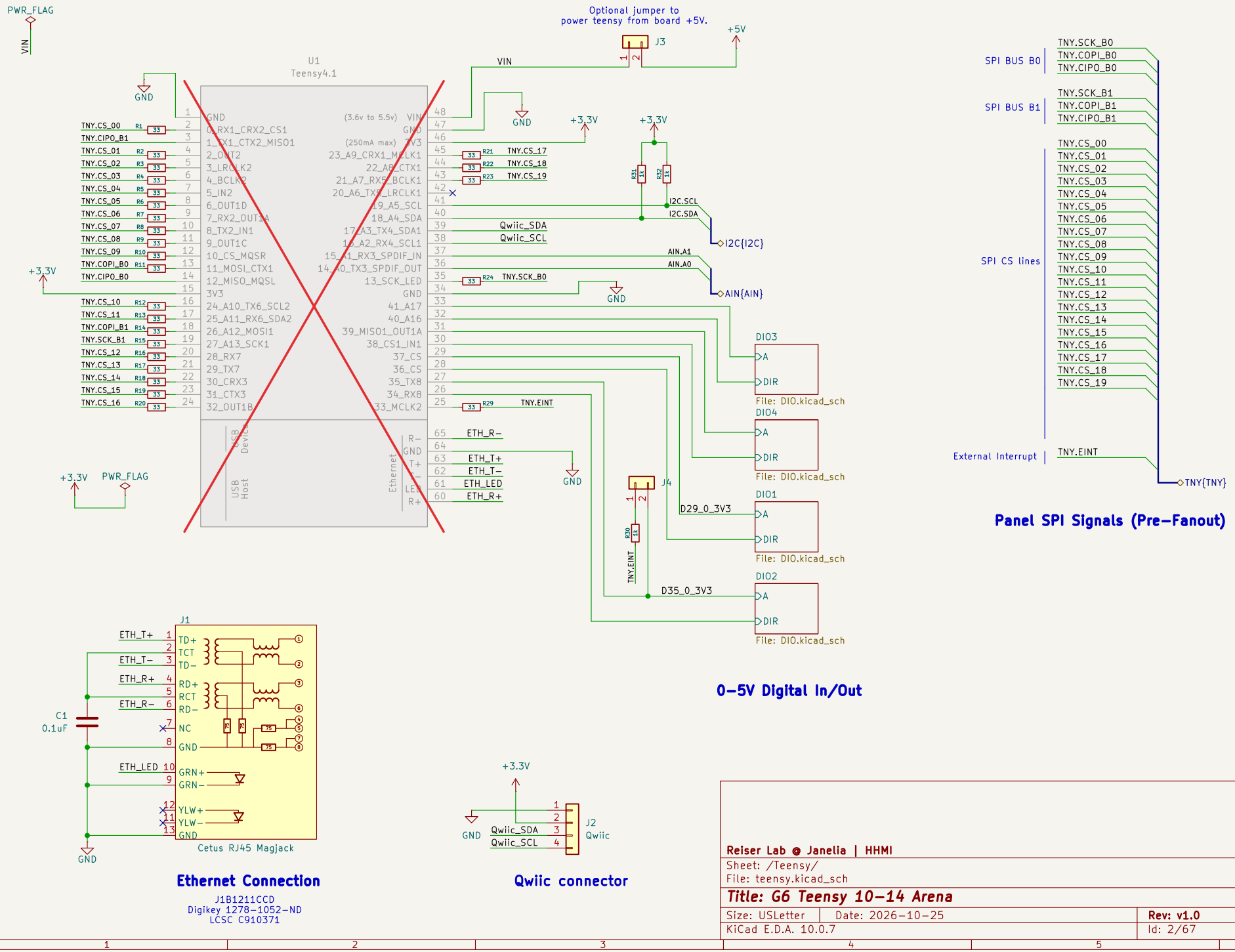


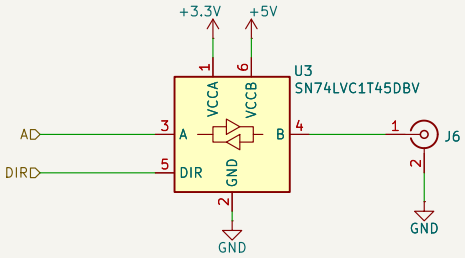

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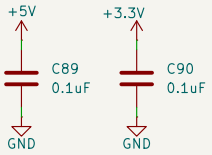
Sheet: /
File: arena_10-14.kicad_sch

***Title: G6 Teensy 10-14 Arena***

| Size: USLetter | Date: 2026-10-25 | **Rev: v1.0** |
|---|---|---|
| KiCad E.D.A. 10.0.7 | | Id: 1/67 |


PWR_FLAG
VIN
Optional jumper to
power teensy from board +5V.
J3
+5V
U1
Teensy4.1
GND
+3.3V
TNY.CS_00
TNY.CIPO_B1
TNY.CS_01
TNY.CS_02
TNY.CS_03
TNY.CS_04
TNY.CS_05
TNY.CS_06
TNY.CS_07
TNY.CS_08
TNY.CS_09
TNY.COPI_B0
TNY.CIPO_B0
TNY.CS_10
TNY.CS_11
TNY.COPI_B1
TNY.SCK_B1
TNY.CS_12
TNY.CS_13
TNY.CS_14
TNY.CS_15
TNY.CS_16
TNY.CS_17
TNY.CS_18
TNY.CS_19
TNY.SCK_B0
TNY.EINT
I2C.SCL
I2C.SDA
Qwiic_SDA
Qwiic_SCL
I2C{I2C}
AIN.A1
AIN.A0
AIN{AIN}
ETH_R−
ETH_T+
ETH_T−
ETH_LED
ETH_R+
D29_0_3V3
D35_0_3V3
DIO3
DIO4
DIO1
DIO2
A
DIR
File: DIO.kicad_sch
J4
SPI BUS B0
SPI BUS B1
SPI CS lines
External Interrupt
TNY{TNY}
Panel SPI Signals (Pre−Fanout)
0−5V Digital In/Out
J1
TD+
TCT
TD−
RD+
RCT
RD−
NC
GND
GRN+
GRN−
YLW+
YLW−
C1
0.1uF
Cetus RJ45 Magjack
Ethernet Connection
J1B1211CCD
Digikey 1278−1052−ND
LCSC C910371
J2
Qwiic
Qwiic connector
Reiser Lab @ Janelia | HHMI
Sheet: /Teensy/
File: teensy.kicad_sch
Title: G6 Teensy 10−14 Arena
Size: USLetter
Date: 2026−10−25
Rev: v1.0
KiCad E.D.A. 10.0.7
Id: 2/67



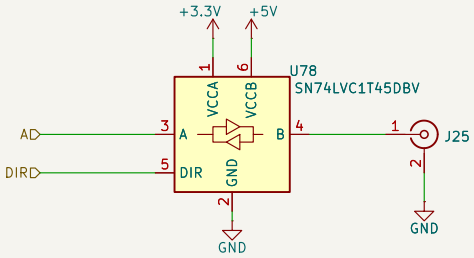

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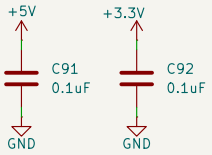
Sheet: /Teensy/DIO2/

File: DIO.kicad_sch

**Title: G6 Teensy 10-14 Arena**

| Size: A4 | Date: 2026-10-25 | **Rev: v1.0** |
|---|---|---|
| KiCad E.D.A. 10.0.7 | | Id: 49/67 |



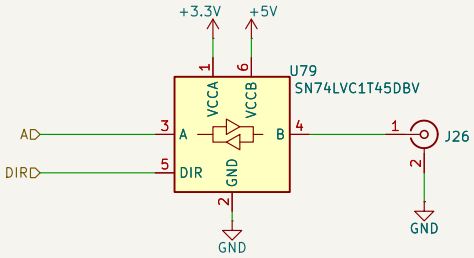



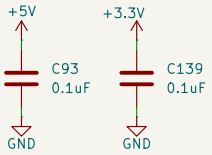

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Sheet: /Teensy/DIO1/
File: DIO.kicad_sch

***Title: G6 Teensy 10-14 Arena***

| Size: A4 | Date: 2026-10-25 | **Rev: v1.0** |
|---|---|---|
| KiCad E.D.A. 10.0.7 | | Id: 50/67 |



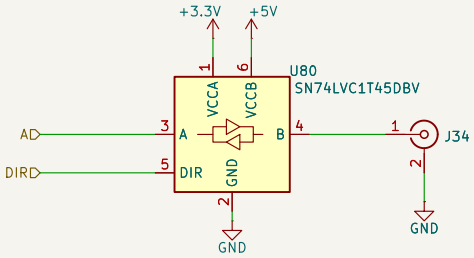

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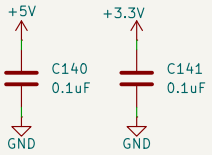
Sheet: /Teensy/DIO4/

File: DIO.kicad_sch

**Title: G6 Teensy 10-14 Arena**

| Size: A4 | Date: 2026-10-25 | **Rev: v1.0** |
|---|---|---|
| KiCad E.D.A. 10.0.7 | | Id: 63/67 |




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Sheet: /Teensy/DIO3/
File: DIO.kicad_sch

***Title: G6 Teensy 10-14 Arena***

| Size: A4 | Date: 2026-10-25 | **Rev: v1.0** |
|---|---|---|
| KiCad E.D.A. 10.0.7 | | Id: 64/67 |

## Singals to/from Teensy

SPI BUS B0
- TNY.SCK_B0
- TNY.COPI_B0
- TNY.CIPO_B0

SPI BUS B1
- TNY.SCK_B1
- TNY.COPI_B1
- TNY.CIPO_B1

SPI CS lines
- TNY.CS_00
- TNY.CS_01
- TNY.CS_02
- TNY.CS_03
- TNY.CS_04
- TNY.CS_05
- TNY.CS_06
- TNY.CS_07
- TNY.CS_08
- TNY.CS_09
- TNY.CS_10
- TNY.CS_11
- TNY.CS_12
- TNY.CS_13
- TNY.CS_14
- TNY.CS_15
- TNY.CS_16
- TNY.CS_17
- TNY.CS_18
- TNY.CS_19

External Interrupt
- TNY.EINT

TNY{TNY}

## Pre-fanout signals

**SCK B0 fan out to P1–P6**
TNY.SCK_B0 → IN; OUT_1 PAN.SCK_B0_P1; OUT_2 PAN.SCK_B0_P2; OUT_3 PAN.SCK_B0_P3; OUT_4 PAN.SCK_B0_P4; OUT_5 PAN.SCK_B0_P5; OUT_6
File: fan_out_by_5x.kicad_sch

**COPI B0 fan out to P1–P6**
TNY.COPI_B0 → IN; OUT_1 PAN.COPI_B0_P1; OUT_2 PAN.COPI_B0_P2; OUT_3 PAN.COPI_B0_P3; OUT_4 PAN.COPI_B0_P4; OUT_5 PAN.COPI_B0_P5; OUT_6
File: fan_out_by_5x.kicad_sch

**SCK B1 fan out to P6–P10**
TNY.SCK_B1 → IN; OUT_1 PAN.SCK_B1_P6; OUT_2 PAN.SCK_B1_P7; OUT_3 PAN.SCK_B1_P8; OUT_4 PAN.SCK_B1_P9; OUT_5 PAN.SCK_B1_P10; OUT_6
File: fan_out_by_5x.kicad_sch

**COPI B1 fan out to P6–P10**
TNY.COPI_B1 → IN; OUT_1 PAN.COPI_B1_P6; OUT_2 PAN.COPI_B1_P7; OUT_3 PAN.COPI_B1_P8; OUT_4 PAN.COPI_B1_P9; OUT_5 PAN.COPI_B1_P10; OUT_6
File: fan_out_by_5x.kicad_sch

**EINT fan out to P1–P5**
TNY.EINT → IN; OUT_1 PAN.EINT_P1; OUT_2 PAN.EINT_P2; OUT_3 PAN.EINT_P3; OUT_4 PAN.EINT_P4; OUT_5 PAN.EINT_P5; OUT_6
File: fan_out_by_5x.kicad_sch

**EINT fan out to P6–P10**
IN; OUT_1 PAN.EINT_P6; OUT_2 PAN.EINT_P7; OUT_3 PAN.EINT_P8; OUT_4 PAN.EINT_P9; OUT_5 PAN.EINT_P10; OUT_6
File: fan_out_by_5x.kicad_sch

**CS_00 fan out to P1, P6**
TNY.CS_00 → IN; OUT_1 PAN.CS_00_P6; OUT_2 PAN.CS_00_P1
File: fan_out_by_2x.kicad_sch

**CS_01 fan out to P1, P6**
TNY.CS_01 → IN; OUT_1 PAN.CS_01_P6; OUT_2 PAN.CS_01_P1
File: fan_out_by_2x.kicad_sch

**CS_02 fan out to P1, P6**
TNY.CS_02 → IN; OUT_1 PAN.CS_02_P6; OUT_2 PAN.CS_02_P1
File: fan_out_by_2x.kicad_sch

**CS_03 fan out to P1, P6**
TNY.CS_03 → IN; OUT_1 PAN.CS_03_P6; OUT_2 PAN.CS_03_P1
File: fan_out_by_2x.kicad_sch

**CS_04 fan out to P2, P7**
TNY.CS_04 → IN; OUT_1 PAN.CS_04_P7; OUT_2 PAN.CS_04_P2
File: fan_out_by_2x.kicad_sch

**CS_05 fan out to P2, P7**
TNY.CS_05 → IN; OUT_1 PAN.CS_05_P7; OUT_2 PAN.CS_05_P2
File: fan_out_by_2x.kicad_sch

**CS_06 fan out to P2, P7**
TNY.CS_06 → IN; OUT_1 PAN.CS_06_P7; OUT_2 PAN.CS_06_P2
File: fan_out_by_2x.kicad_sch

**CS_07 fan out to P2, P7**
TNY.CS_07 → IN; OUT_1 PAN.CS_07_P7; OUT_2 PAN.CS_07_P2
File: fan_out_by_2x.kicad_sch

**CS_08 fan out to P3, P8**
TNY.CS_08 → IN; OUT_1 PAN.CS_08_P8; OUT_2 PAN.CS_08_P3
File: fan_out_by_2x.kicad_sch

**CS_09 fan out to P3, P8**
TNY.CS_09 → IN; OUT_1 PAN.CS_09_P8; OUT_2 PAN.CS_09_P3
File: fan_out_by_2x.kicad_sch

**CS_10 fan out to P3, P8**
TNY.CS_10 → IN; OUT_1 PAN.CS_10_P8; OUT_2 PAN.CS_10_P3
File: fan_out_by_2x.kicad_sch

**CS_11 fan out to P3, P8**
TNY.CS_11 → IN; OUT_1 PAN.CS_11_P8; OUT_2 PAN.CS_11_P3
File: fan_out_by_2x.kicad_sch

**CS_12 fan out to P4, P9**
TNY.CS_12 → IN; OUT_1 PAN.CS_12_P9; OUT_2 PAN.CS_12_P4
File: fan_out_by_2x.kicad_sch

**CS_13 fan out to P4, P9**
TNY.CS_13 → IN; OUT_1 PAN.CS_13_P9; OUT_2 PAN.CS_13_P4
File: fan_out_by_2x.kicad_sch

**CS_14 fan out to P4, P9**
TNY.CS_14 → IN; OUT_1 PAN.CS_14_P9; OUT_2 PAN.CS_14_P4
File: fan_out_by_2x.kicad_sch

**CS_15 fan out to P4, P9**
TNY.CS_15 → IN; OUT_1 PAN.CS_15_P9; OUT_2 PAN.CS_15_P4
File: fan_out_by_2x.kicad_sch

**CS_16 fan out to P5, P10**
TNY.CS_16 → IN; OUT_1 PAN.CS_16_P10; OUT_2 PAN.CS_16_P5
File: fan_out_by_2x.kicad_sch

**CS_17 fan out to P5, P10**
TNY.CS_17 → IN; OUT_1 PAN.CS_17_P10; OUT_2 PAN.CS_17_P5
File: fan_out_by_2x.kicad_sch

**CS_18 fan out to P5, P10**
TNY.CS_18 → IN; OUT_1 PAN.CS_18_P10; OUT_2 PAN.CS_18_P5
File: fan_out_by_2x.kicad_sch

**CS_19 fan out to P5, P10**
TNY.CS_19 → IN; OUT_1 PAN.CS_19_P10; OUT_2 PAN.CS_19_P5
File: fan_out_by_2x.kicad_sch

**SPI Bus B0**
- PAN.SCK_B0_P1
- PAN.SCK_B0_P2
- PAN.SCK_B0_P3
- PAN.SCK_B0_P4
- PAN.SCK_B0_P5
- PAN.COPI_B0_P1
- PAN.COPI_B0_P2
- PAN.COPI_B0_P3
- PAN.COPI_B0_P4
- PAN.COPI_B0_P5
- PAN.CIPO_B0_P1
- PAN.CIPO_B0_P2
- PAN.CIPO_B0_P3
- PAN.CIPO_B0_P4
- PAN.CIPO_B0_P5

**SPI Bus B1**
- PAN.SCK_B1_P6
- PAN.SCK_B1_P7
- PAN.SCK_B1_P8
- PAN.SCK_B1_P9
- PAN.SCK_B1_P10
- PAN.COPI_B1_P6
- PAN.COPI_B1_P7
- PAN.COPI_B1_P8
- PAN.COPI_B1_P9
- PAN.COPI_B1_P10
- PAN.CIPO_B1_P6
- PAN.CIPO_B1_P7
- PAN.CIPO_B1_P8
- PAN.CIPO_B1_P9
- PAN.CIPO_B1_P10

**Chip Select P1–P10**
- PAN.CS_00_P1
- PAN.CS_01_P1
- PAN.CS_02_P1
- PAN.CS_03_P1
- PAN.CS_04_P2
- PAN.CS_05_P2
- PAN.CS_06_P2
- PAN.CS_07_P2
- PAN.CS_08_P3
- PAN.CS_09_P3
- PAN.CS_10_P3
- PAN.CS_11_P3
- PAN.CS_12_P4
- PAN.CS_13_P4
- PAN.CS_14_P4
- PAN.CS_15_P4
- PAN.CS_16_P5
- PAN.CS_17_P5
- PAN.CS_18_P5
- PAN.CS_19_P5
- PAN.CS_00_P6
- PAN.CS_01_P6
- PAN.CS_02_P6
- PAN.CS_03_P6
- PAN.CS_04_P7
- PAN.CS_05_P7
- PAN.CS_06_P7
- PAN.CS_07_P7
- PAN.CS_08_P8
- PAN.CS_09_P8
- PAN.CS_10_P8
- PAN.CS_11_P8
- PAN.CS_12_P9
- PAN.CS_13_P9
- PAN.CS_14_P9
- PAN.CS_15_P9
- PAN.CS_16_P10
- PAN.CS_17_P10
- PAN.CS_18_P10
- PAN.CS_19_P10

**EINT**
- PAN.EINT_P1
- PAN.EINT_P2
- PAN.EINT_P3
- PAN.EINT_P4
- PAN.EINT_P5
- PAN.EINT_P6
- PAN.EINT_P7
- PAN.EINT_P8
- PAN.EINT_P9
- PAN.EINT_P10

PAN{PAN}

## Post-Fanout Signals

**CIPO B0 Enable for P1**
CS0 PAN.CS_00_P1; CS1 PAN.CS_01_P1; CS2 PAN.CS_02_P1; CS3 PAN.CS_03_P1; CIPO_OUT; CIPO_IN PAN.CIPO_B0_P1
File: cipo_enable.kicad_sch
TNY.CIPO_B0

**CIPO B0 Enable for P2**
CS0 PAN.CS_04_P2; CS1 PAN.CS_05_P2; CS2 PAN.CS_06_P2; CS3 PAN.CS_07_P2; CIPO_OUT; CIPO_IN PAN.CIPO_B0_P2
File: cipo_enable.kicad_sch

**CIPO B0 Enable for P3**
CS0 PAN.CS_08_P3; CS1 PAN.CS_09_P3; CS2 PAN.CS_10_P3; CS3 PAN.CS_11_P3; CIPO_OUT; CIPO_IN PAN.CIPO_B0_P3
File: cipo_enable.kicad_sch

**CIPO B0 Enable for P4**
CS0 PAN.CS_12_P4; CS1 PAN.CS_13_P4; CS2 PAN.CS_14_P4; CS3 PAN.CS_15_P4; CIPO_OUT; CIPO_IN PAN.CIPO_B0_P4
File: cipo_enable.kicad_sch

**CIPO B0 Enable for P5**
CS0 PAN.CS_16_P5; CS1 PAN.CS_17_P5; CS2 PAN.CS_18_P5; CS3 PAN.CS_19_P5; CIPO_OUT; CIPO_IN PAN.CIPO_B0_P5
File: cipo_enable.kicad_sch

**CIPO B1 Enable for P6**
CS0 PAN.CS_00_P6; CS1 PAN.CS_01_P6; CS2 PAN.CS_02_P6; CS3 PAN.CS_03_P6; CIPO_OUT; CIPO_IN PAN.CIPO_B1_P6
File: cipo_enable.kicad_sch
TNY.CIPO_B1

**CIPO B1 Enable for P7**
CS0 PAN.CS_04_P7; CS1 PAN.CS_05_P7; CS2 PAN.CS_06_P7; CS3 PAN.CS_07_P7; CIPO_OUT; CIPO_IN PAN.CIPO_B1_P7
File: cipo_enable.kicad_sch

**CIPO B1 Enable for P8**
CS0 PAN.CS_08_P8; CS1 PAN.CS_09_P8; CS2 PAN.CS_10_P8; CS3 PAN.CS_11_P8; CIPO_OUT; CIPO_IN PAN.CIPO_B1_P8
File: cipo_enable.kicad_sch

**CIPO B1 Enable for P9**
CS0 PAN.CS_12_P9; CS1 PAN.CS_13_P9; CS2 PAN.CS_14_P9; CS3 PAN.CS_15_P9; CIPO_OUT; CIPO_IN PAN.CIPO_B1_P9
File: cipo_enable.kicad_sch

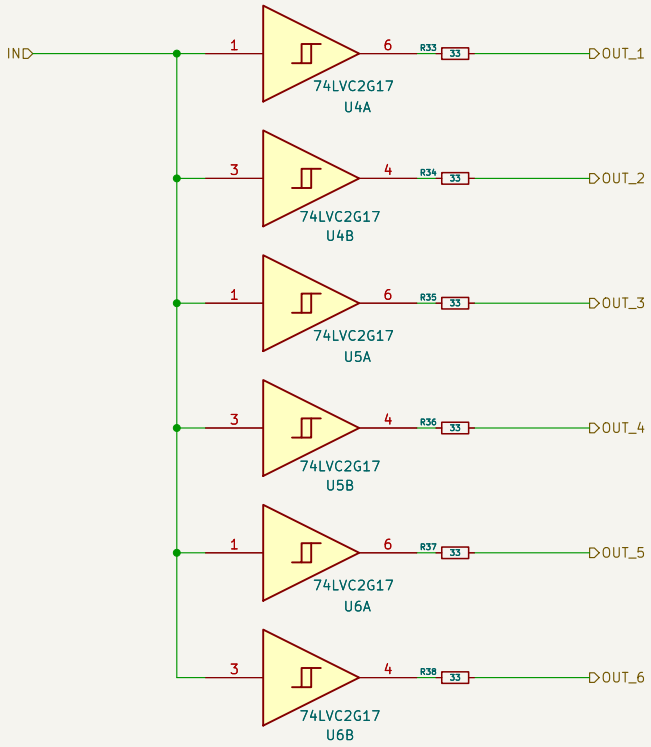
**CIPO B1 Enable for P10**
CS0 PAN.CS_16_P10; CS1 PAN.CS_17_P10; CS2 PAN.CS_18_P10; CS3 PAN.CS_19_P10; CIPO_OUT; CIPO_IN PAN.CIPO_B1_P10
File: cipo_enable.kicad_sch

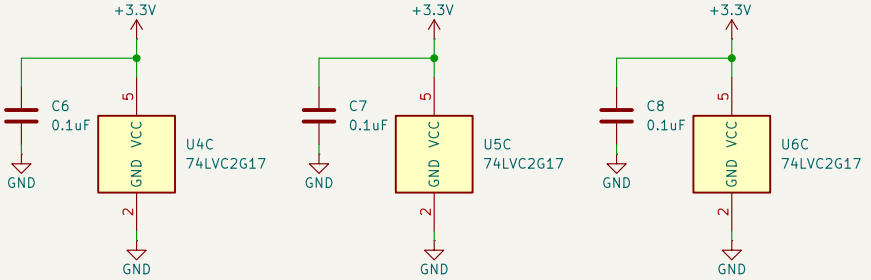
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Sheet: /Fan Out/
File: fan_out.kicad_sch

**Title: G6 Teensy 10–14 Arena**

Size: USLedger | Date: 2026-10-25 | **Rev: v1.0**
KiCad E.D.A. 10.0.7 | Id: 3/67



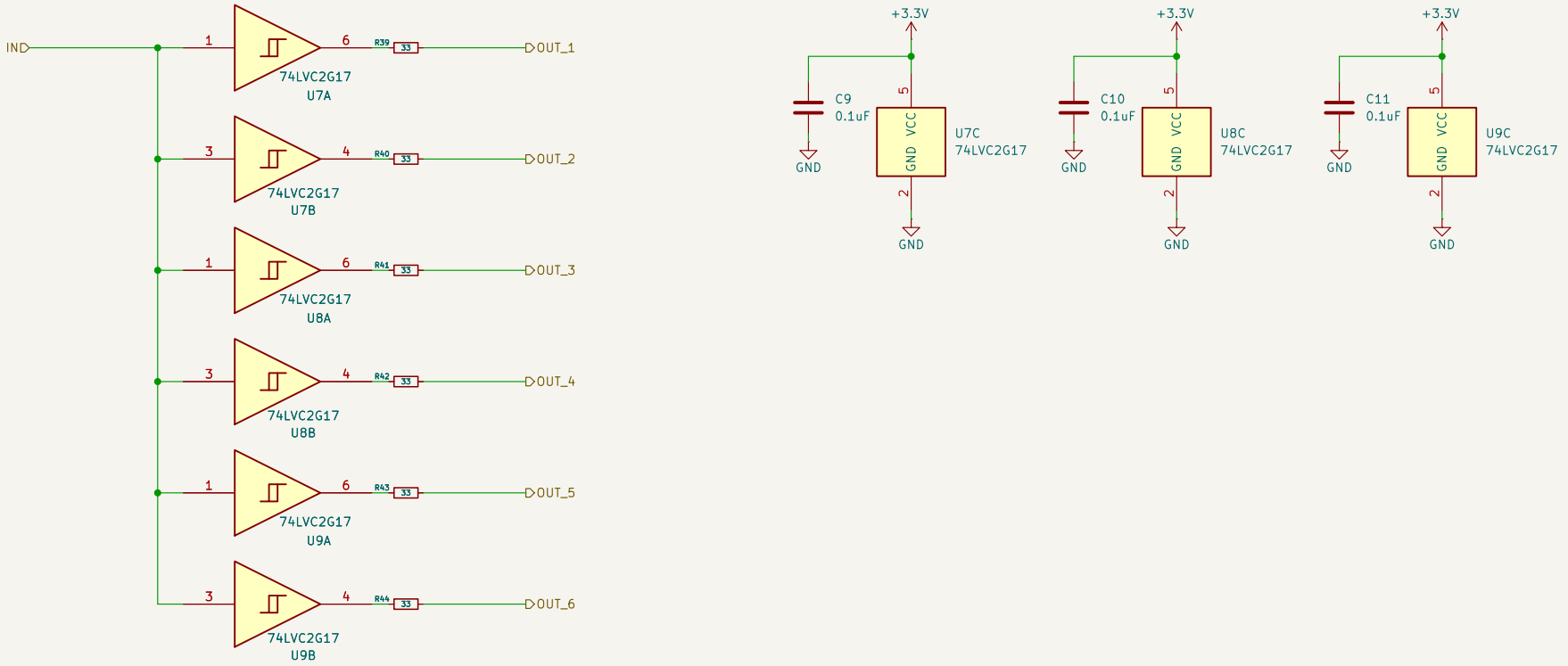




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Sheet: /Fan Out/SCK B0 fan out to P1–P6/
File: fan_out_by_5x.kicad_sch

***Title: G6 Teensy 10–14 Arena***

| Size: USLetter | Date: 2026-10-25 | **Rev: v1.0** |
|---|---|---|
| KiCad E.D.A. 10.0.7 | | Id: 4/67 |



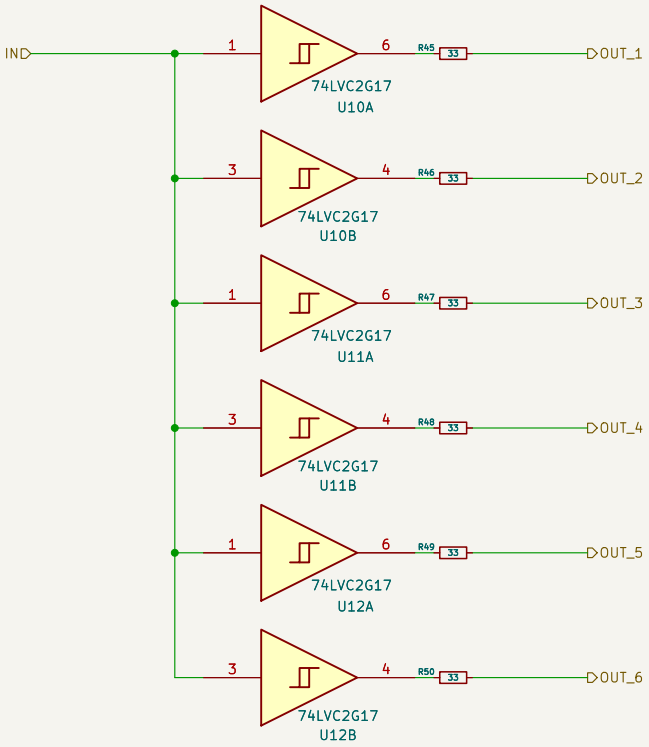

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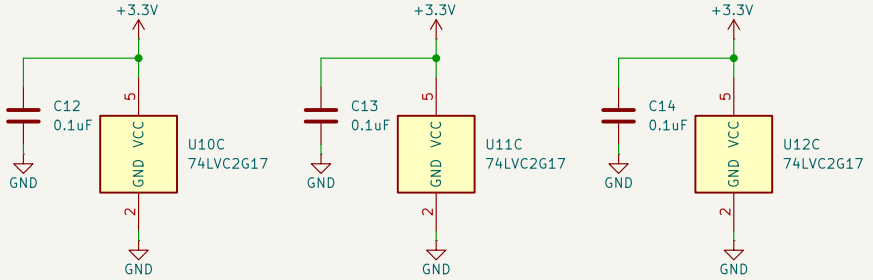
Sheet: /Fan Out/COPI B0 fan out to P1–P6/

File: fan_out_by_5x.kicad_sch

***Title: G6 Teensy 10–14 Arena***

| Size: USLetter | Date: 2026-10-25 | **Rev: v1.0** |
|---|---|---|
| KiCad E.D.A. 10.0.7 | | Id: 5/67 |



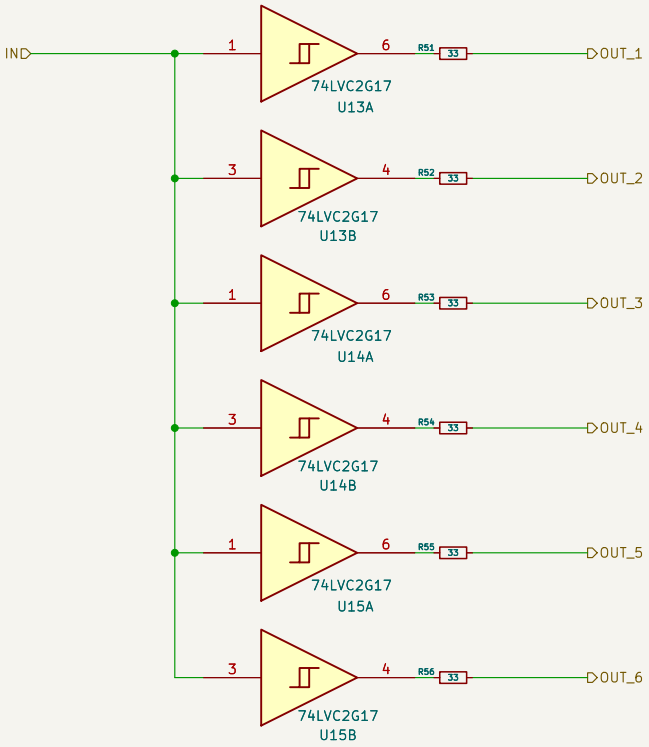



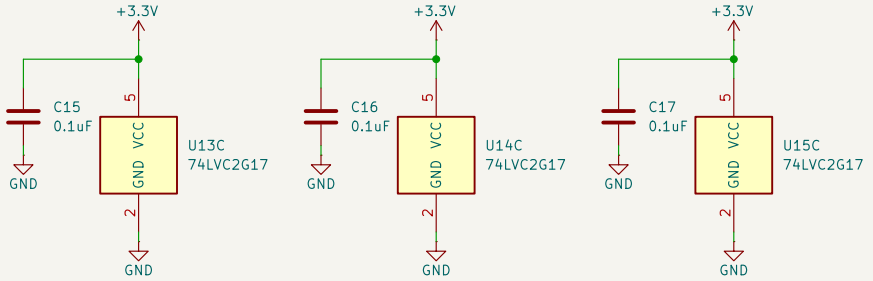

**Reiser Lab @ Janelia | HHMI**

Sheet: /Fan Out/SCK B1 fan out to P6–P10/
File: fan_out_by_5x.kicad_sch

***Title: G6 Teensy 10–14 Arena***

| Size: USLetter | Date: 2026-10-25 | **Rev: v1.0** |
|---|---|---|
| KiCad E.D.A. 10.0.7 | | Id: 6/67 |



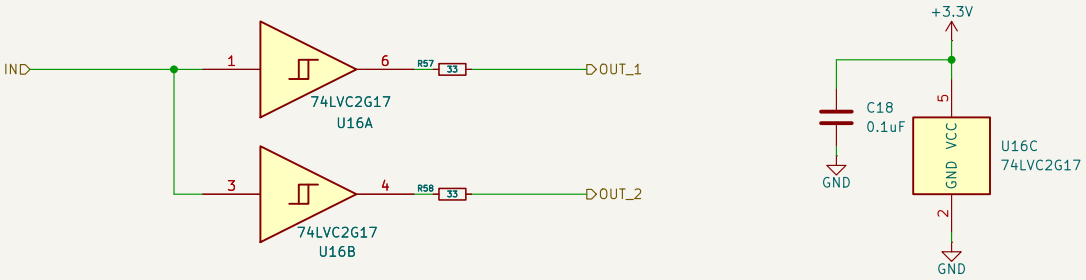




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Sheet: /Fan Out/COPI B1 fan out to P6-P10/
File: fan_out_by_5x.kicad_sch

***Title: G6 Teensy 10-14 Arena***

| Size: USLetter | Date: 2026-10-25 | **Rev: v1.0** |
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| KiCad E.D.A. 10.0.7 | | Id: 7/67 |







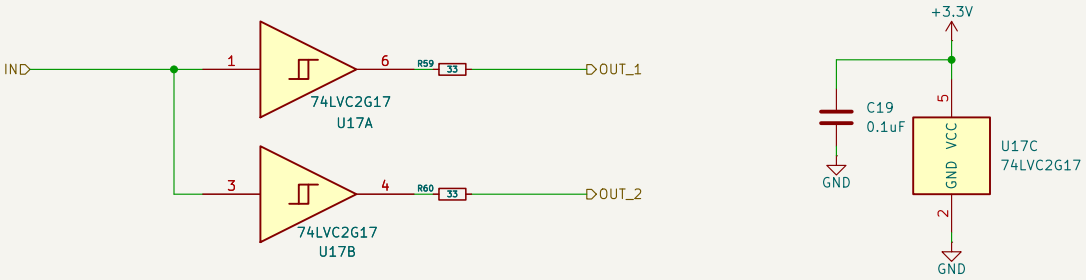
Reiser Lab @ Janelia | HHMI

Sheet: /Fan Out/CS_00 fan out to P1, P6/

File: fan_out_by_2x.kicad_sch

**Title: G6 Teensy 10-14 Arena**

| Size: USLetter | Date: 2026-10-25 | **Rev: v1.0** |
|---|---|---|
| KiCad E.D.A. 10.0.7 | | Id: 8/67 |





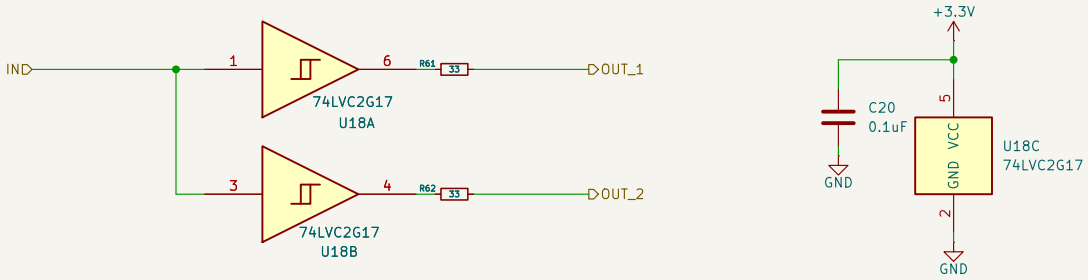


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Sheet: /Fan Out/CS_01 fan out to P1, P6/
File: fan_out_by_2x.kicad_sch

***Title: G6 Teensy 10-14 Arena***

| Size: USLetter | Date: 2026-10-25 | **Rev: v1.0** |
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| KiCad E.D.A. 10.0.7 | | Id: 9/67 |







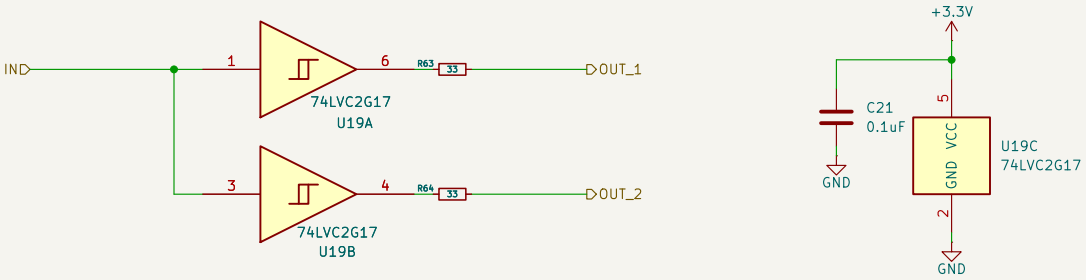
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Sheet: /Fan Out/CS_02 fan out to P1, P6/

File: fan_out_by_2x.kicad_sch

**Title: G6 Teensy 10-14 Arena**

| Size: USLetter | Date: 2026-10-25 | **Rev: v1.0** |
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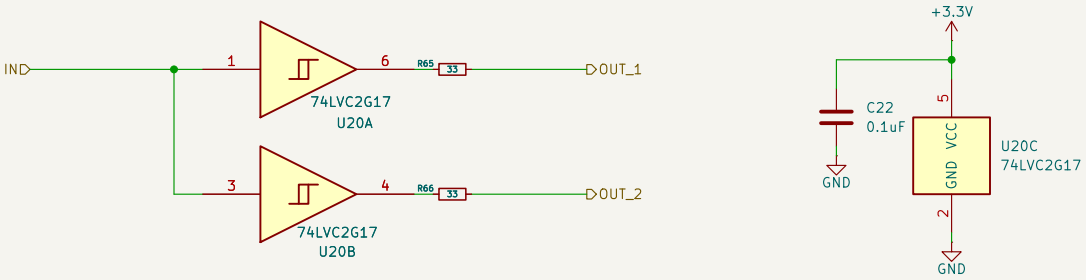
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Sheet: /Fan Out/CS_03 fan out to P1, P6/

File: fan_out_by_2x.kicad_sch

**Title: G6 Teensy 10-14 Arena**

| | | |
|---|---|---|
| Size: USLetter | Date: 2026-10-25 | **Rev: v1.0** |
| KiCad E.D.A. 10.0.7 | | Id: 11/67 |







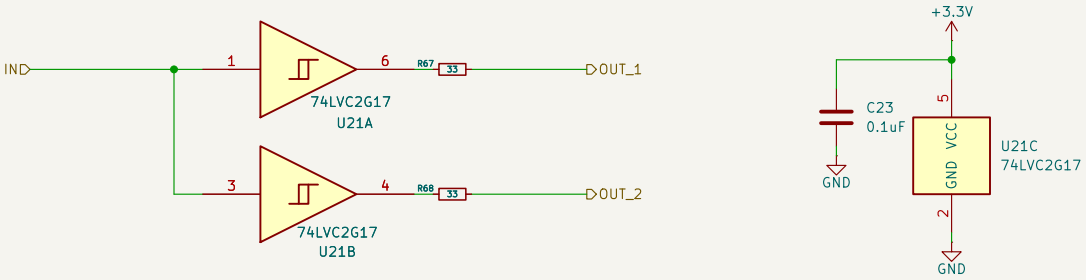
**Reiser Lab @ Janelia | HHMI**

Sheet: /Fan Out/CS_04 fan out to P2, P7/

File: fan_out_by_2x.kicad_sch

***Title: G6 Teensy 10–14 Arena***

| Size: USLetter | Date: 2026-10-25 | **Rev: v1.0** |
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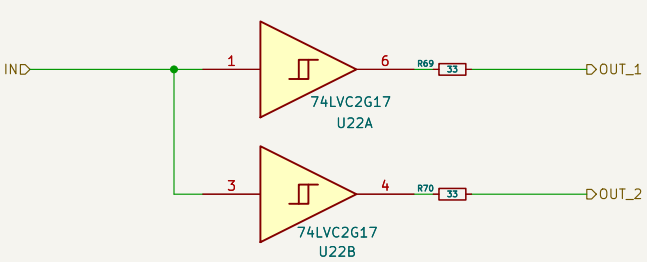
Reiser Lab @ Janelia | HHMI

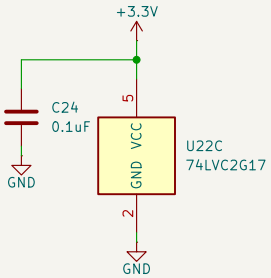
Sheet: /Fan Out/CS_05 fan out to P2, P7/

File: fan_out_by_2x.kicad_sch

**Title: G6 Teensy 10-14 Arena**

| Size: USLetter | Date: 2026-10-25 | **Rev: v1.0** |
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| KiCad E.D.A. 10.0.7 | | Id: 13/67 |



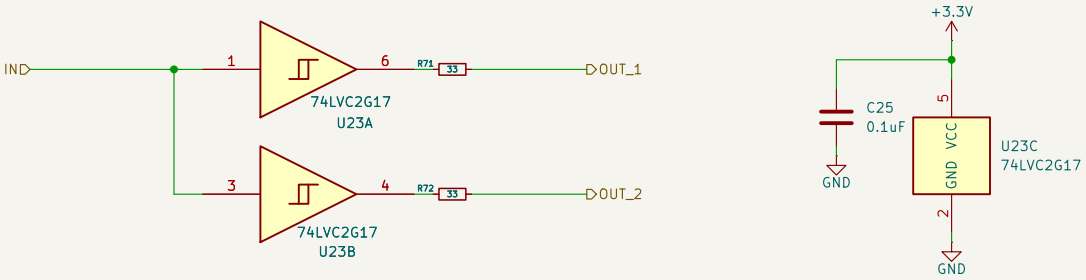

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Sheet: /Fan Out/CS_06 fan out to P2, P7/

File: fan_out_by_2x.kicad_sch

***Title: G6 Teensy 10–14 Arena***

| Size: USLetter | Date: 2026-10-25 | **Rev: v1.0** |
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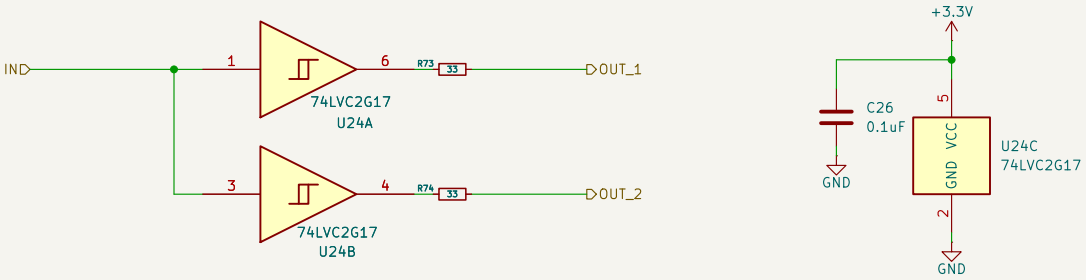


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Sheet: /Fan Out/CS_07 fan out to P2, P7/
File: fan_out_by_2x.kicad_sch

***Title: G6 Teensy 10-14 Arena***

| Size: USLetter | Date: 2026-10-25 | **Rev: v1.0** |
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| KiCad E.D.A. 10.0.7 | | Id: 15/67 |





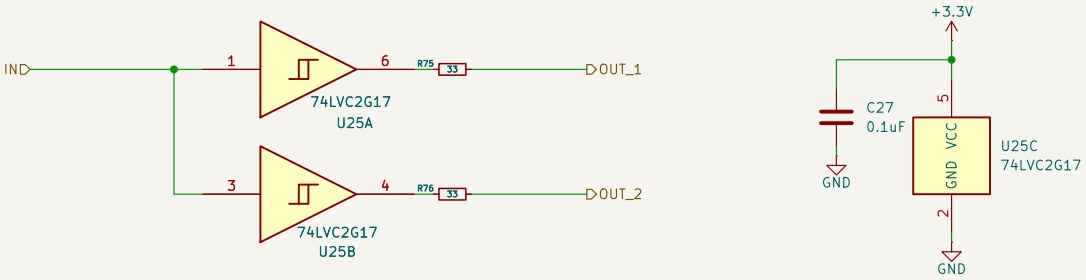


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Sheet: /Fan Out/CS_08 fan out to P3, P8/
File: fan_out_by_2x.kicad_sch

***Title: G6 Teensy 10-14 Arena***

| Size: USLetter | Date: 2026-10-25 | **Rev: v1.0** |
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| KiCad E.D.A. 10.0.7 | | Id: 16/67 |





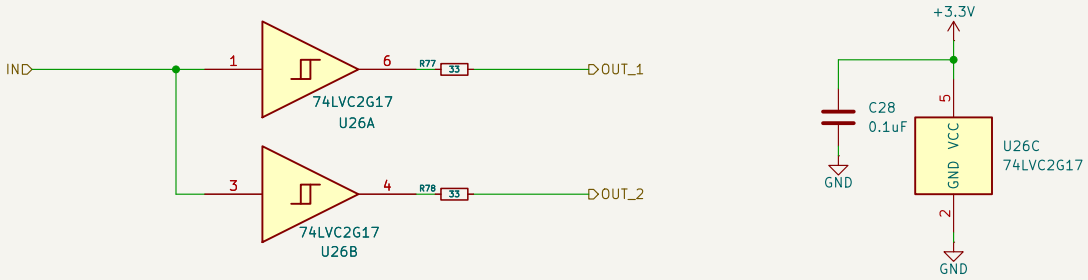


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Sheet: /Fan Out/CS_09 fan out to P3, P8/
File: fan_out_by_2x.kicad_sch

***Title: G6 Teensy 10-14 Arena***

| Size: USLetter | Date: 2026-10-25 | **Rev: v1.0** |
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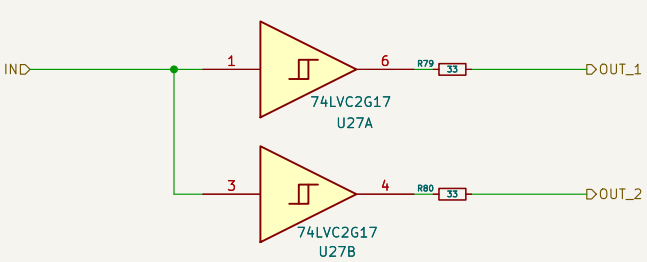


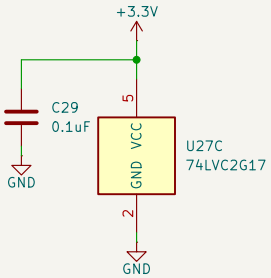
**Reiser Lab @ Janelia | HHMI**

Sheet: /Fan Out/CS_10 fan out to P3, P8/
File: fan_out_by_2x.kicad_sch

***Title: G6 Teensy 10-14 Arena***

| Size: USLetter | Date: 2026-10-25 | **Rev: v1.0** |
|---|---|---|
| KiCad E.D.A. 10.0.7 | | Id: 18/67 |



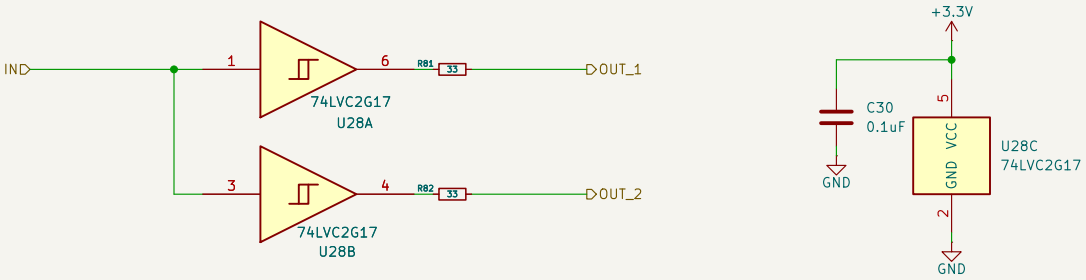




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Sheet: /Fan Out/CS_11 fan out to P3, P8/
File: fan_out_by_2x.kicad_sch

***Title: G6 Teensy 10–14 Arena***

| Size: USLetter | Date: 2026-10-25 | **Rev: v1.0** |
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| KiCad E.D.A. 10.0.7 | | Id: 19/67 |





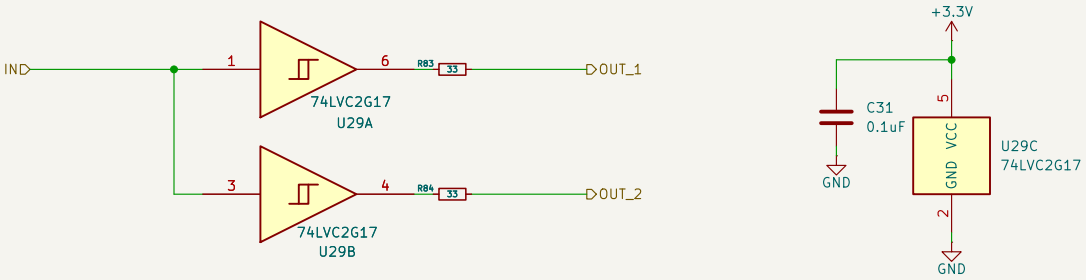


**Reiser Lab @ Janelia | HHMI**

Sheet: /Fan Out/CS_12 fan out to P4, P9/
File: fan_out_by_2x.kicad_sch

***Title: G6 Teensy 10-14 Arena***

| Size: USLetter | Date: 2026-10-25 | **Rev: v1.0** |
|---|---|---|
| KiCad E.D.A. 10.0.7 | | Id: 20/67 |







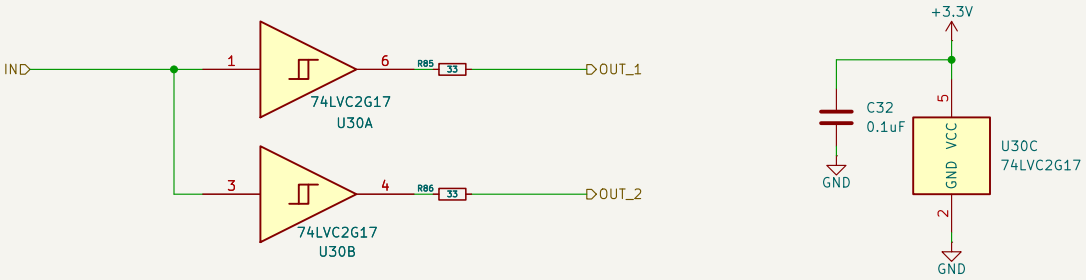
Reiser Lab @ Janelia | HHMI

Sheet: /Fan Out/CS_13 fan out to P4, P9/

File: fan_out_by_2x.kicad_sch

**Title: G6 Teensy 10-14 Arena**

| Size: USLetter | Date: 2026-10-25 | **Rev: v1.0** |
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| KiCad E.D.A. 10.0.7 | | Id: 21/67 |







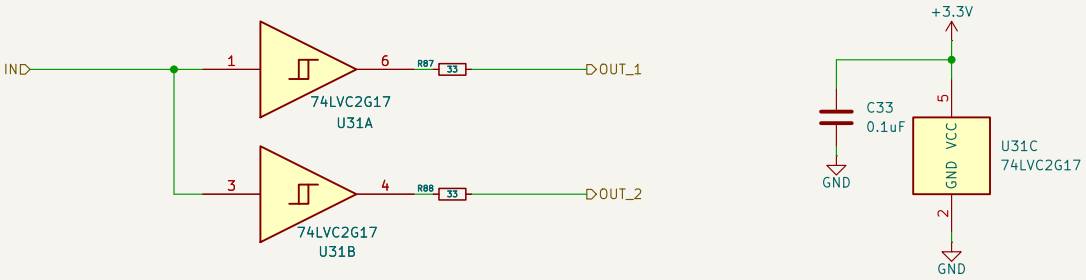
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Sheet: /Fan Out/CS_14 fan out to P4, P9/

File: fan_out_by_2x.kicad_sch

**Title: G6 Teensy 10–14 Arena**

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|---|---|---|
| Size: USLetter | Date: 2026-10-25 | **Rev: v1.0** |
| KiCad E.D.A. 10.0.7 | | Id: 22/67 |





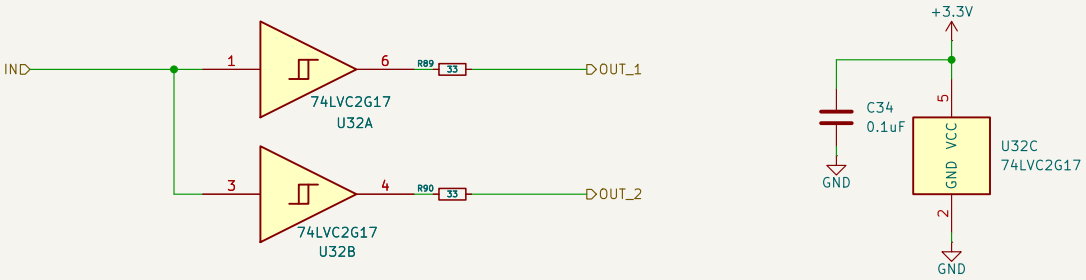


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Sheet: /Fan Out/CS_15 fan out to P4, P9/
File: fan_out_by_2x.kicad_sch

***Title: G6 Teensy 10-14 Arena***

| Size: USLetter | Date: 2026-10-25 | **Rev: v1.0** |
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| KiCad E.D.A. 10.0.7 | | Id: 23/67 |





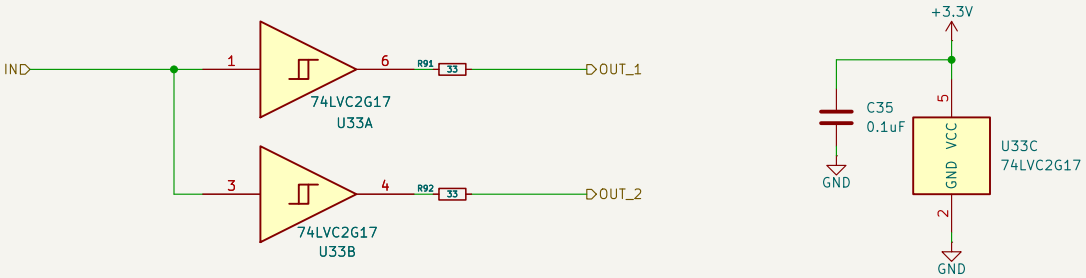


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Sheet: /Fan Out/CS_16 fan out to P5, P10/
File: fan_out_by_2x.kicad_sch

***Title: G6 Teensy 10-14 Arena***

| Size: USLetter | Date: 2026-10-25 | **Rev: v1.0** |
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| KiCad E.D.A. 10.0.7 | | Id: 24/67 |







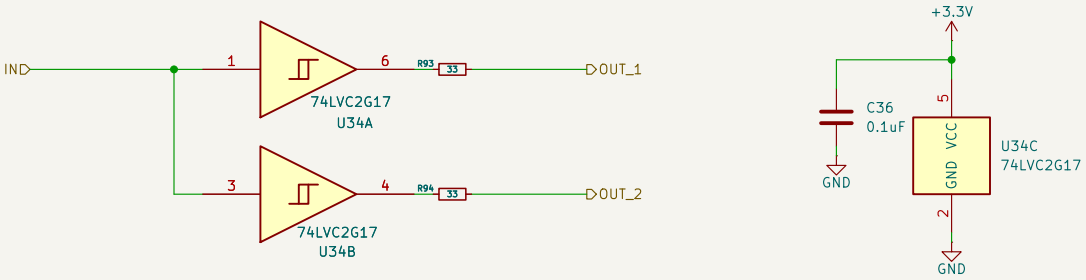
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Sheet: /Fan Out/CS_17 fan out to P5, P10/

File: fan_out_by_2x.kicad_sch

**Title: G6 Teensy 10-14 Arena**

| Size: USLetter | Date: 2026-10-25 | **Rev: v1.0** |
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| KiCad E.D.A. 10.0.7 | | Id: 25/67 |







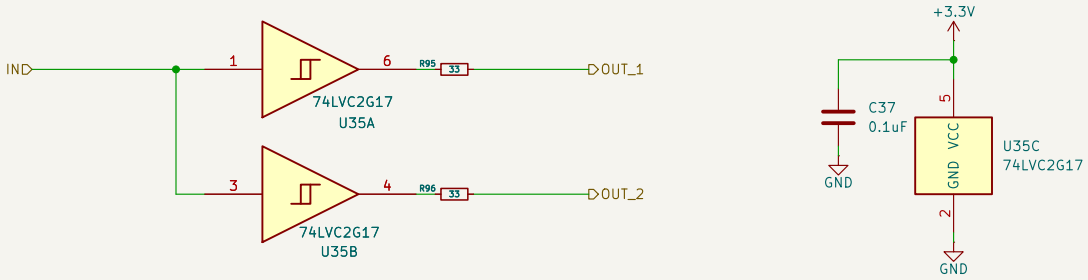
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Sheet: /Fan Out/CS_18 fan out to P5, P10/

File: fan_out_by_2x.kicad_sch

**Title: G6 Teensy 10–14 Arena**

| Size: USLetter | Date: 2026-10-25 | **Rev: v1.0** |
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| KiCad E.D.A. 10.0.7 | | Id: 26/67 |







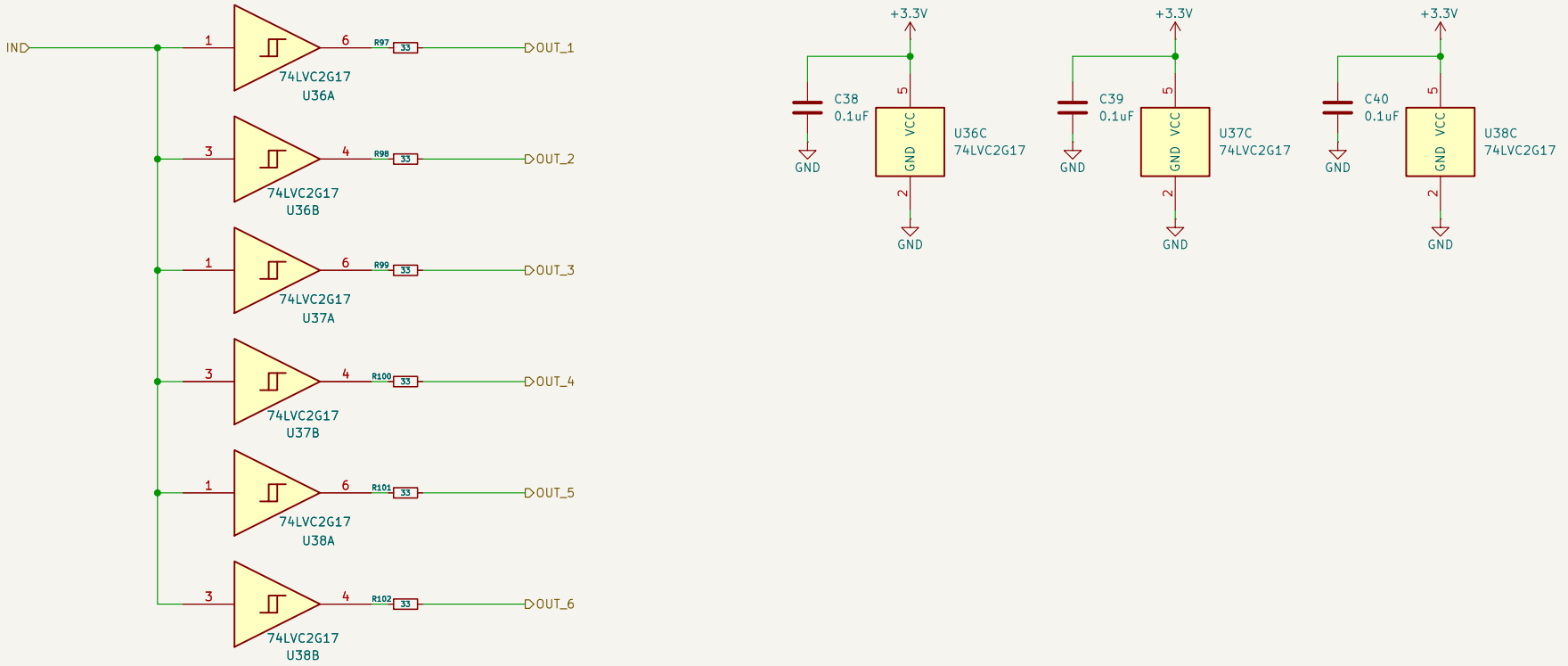
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Sheet: /Fan Out/CS_19 fan out to P5, P10/

File: fan_out_by_2x.kicad_sch

**Title: G6 Teensy 10–14 Arena**

| Size: USLetter | Date: 2026-10-25 | **Rev: v1.0** |
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| KiCad E.D.A. 10.0.7 | | Id: 27/67 |

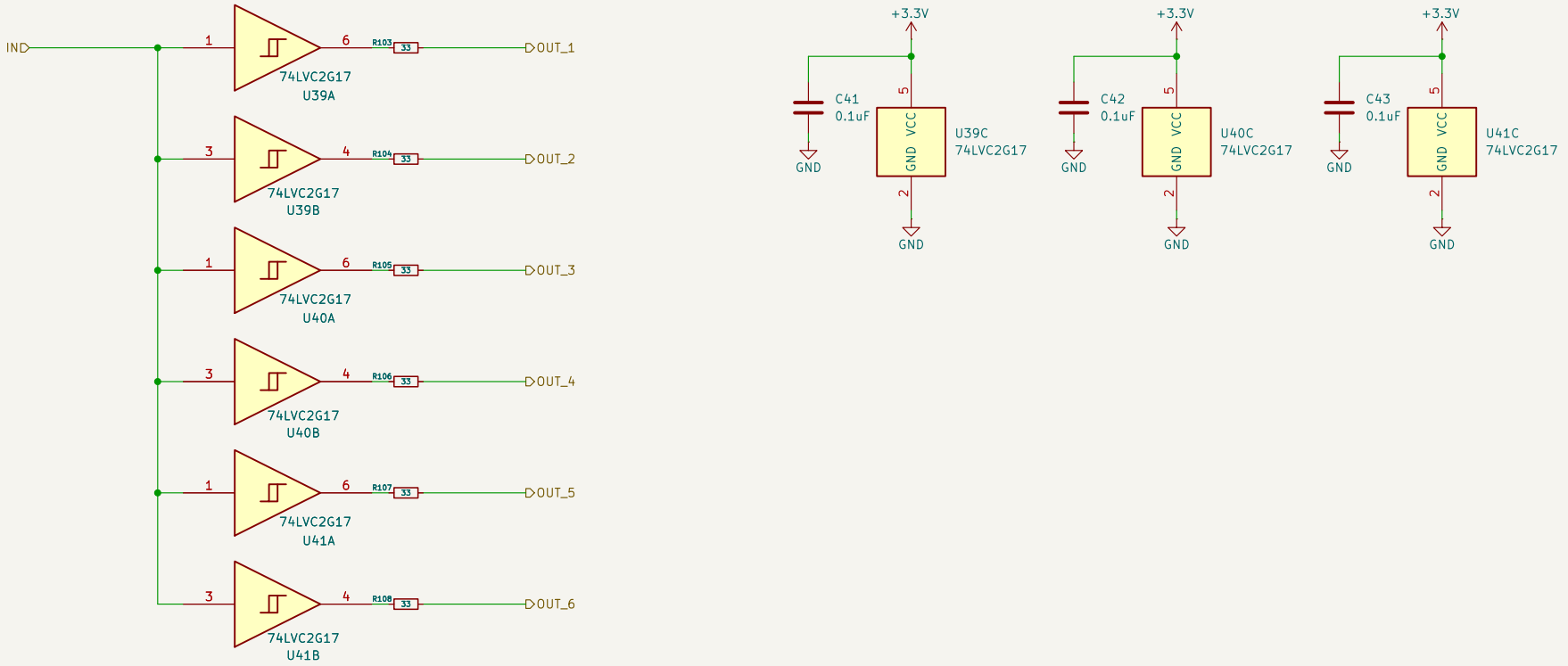



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Sheet: /Fan Out/EINT fan out to P6–P10/
File: fan_out_by_5x.kicad_sch

***Title: G6 Teensy 10–14 Arena***

| Size: USLetter | Date: 2026-10-25 | **Rev: v1.0** |
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| KiCad E.D.A. 10.0.7 | | Id: 28/67 |



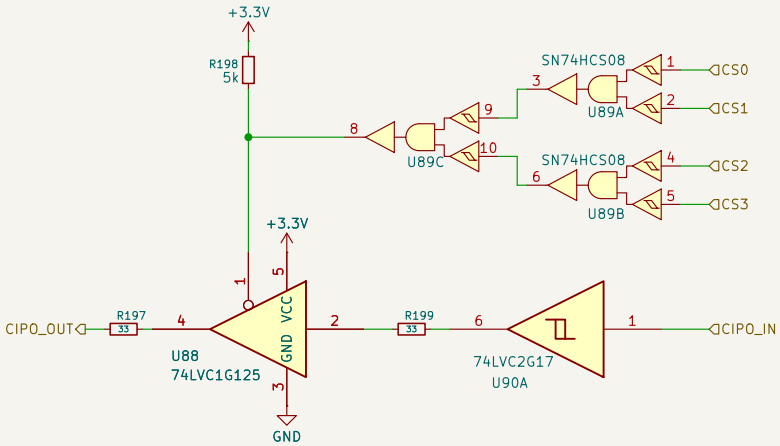

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Sheet: /Fan Out/EINT fan out to P1–P5/

File: fan_out_by_5x.kicad_sch

***Title: G6 Teensy 10–14 Arena***

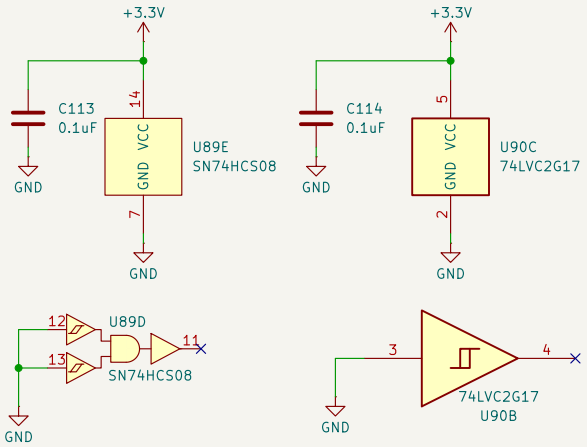
| Size: USLetter | Date: 2026-10-25 | **Rev: v1.0** |
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| KiCad E.D.A. 10.0.7 | | Id: 29/67 |



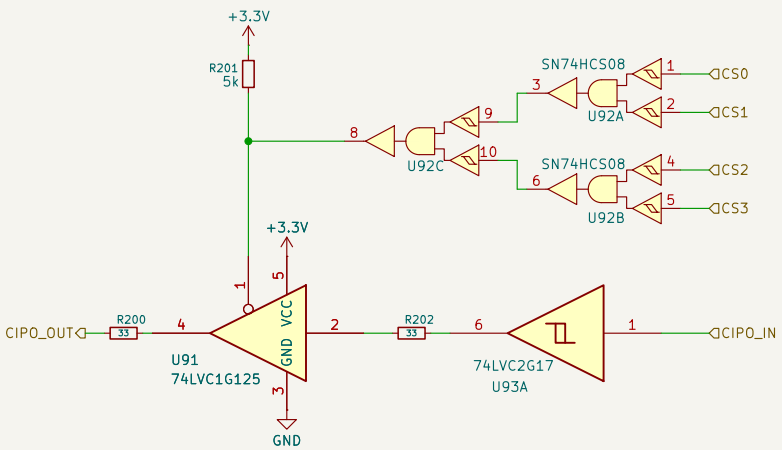




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C10429



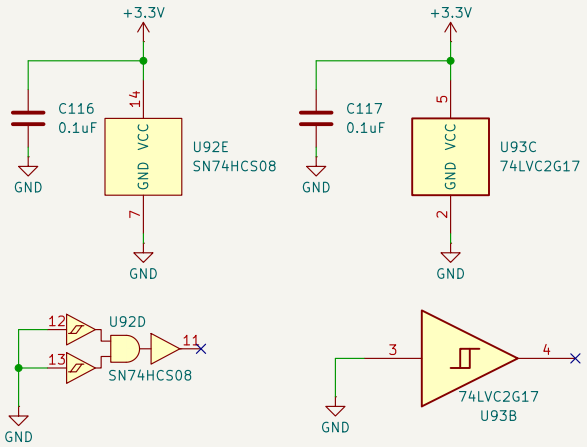

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|---|---|
| **Reiser Lab @ Janelia \| HHMI** | |
| Sheet: /Fan Out/CIPO B0 Enable for P1/ | |
| File: cipo_enable.kicad_sch | |
| ***Title: G6 Teensy 10–14 Arena*** | |
| Size: USLetter \| Date: 2026-10-25 | **Rev: v1.0** |
| KiCad E.D.A. 10.0.7 | Id: 54/67 |



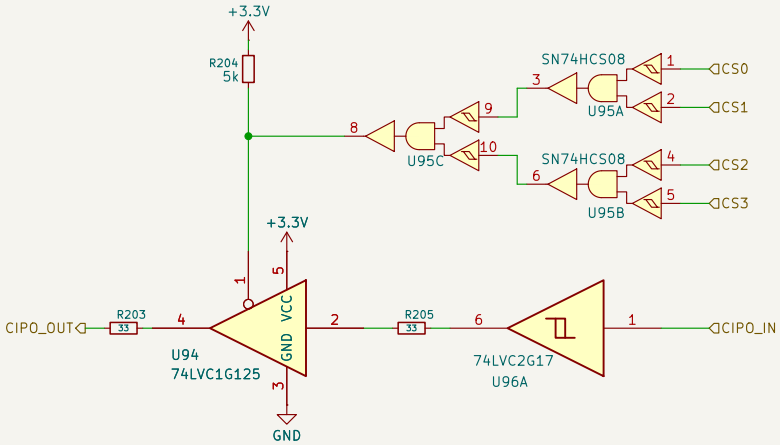


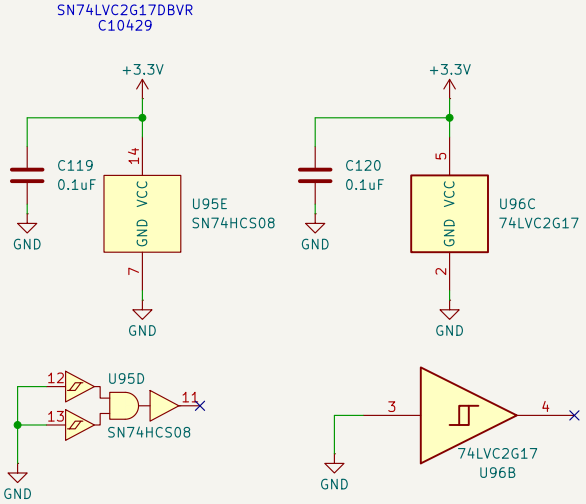


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|---|---|
| **Reiser Lab @ Janelia \| HHMI** | |
| Sheet: /Fan Out/CIPO B0 Enable for P2/ | |
| File: cipo_enable.kicad_sch | |
| ***Title: G6 Teensy 10–14 Arena*** | |
| Size: USLetter \| Date: 2026-10-25 | **Rev: v1.0** |
| KiCad E.D.A. 10.0.7 | Id: 55/67 |





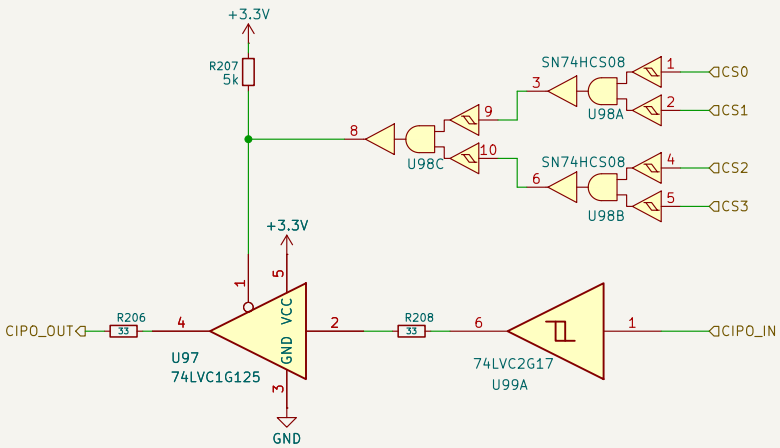


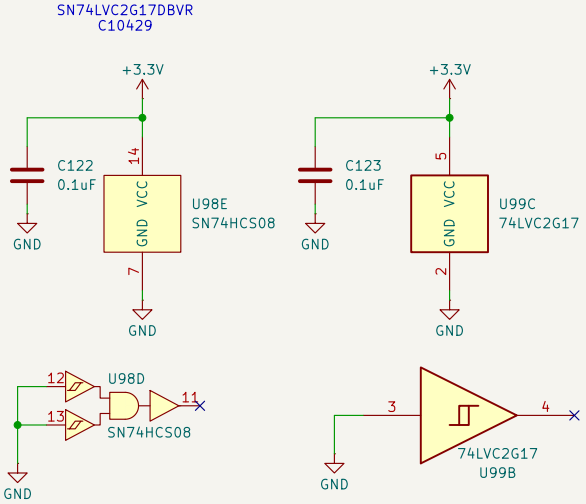
**Reiser Lab @ Janelia | HHMI**

Sheet: /Fan Out/CIPO B0 Enable for P3/
File: cipo_enable.kicad_sch

***Title: G6 Teensy 10–14 Arena***

| Size: USLetter | Date: 2026-10-25 | **Rev: v1.0** |
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| KiCad E.D.A. 10.0.7 | | Id: 56/67 |

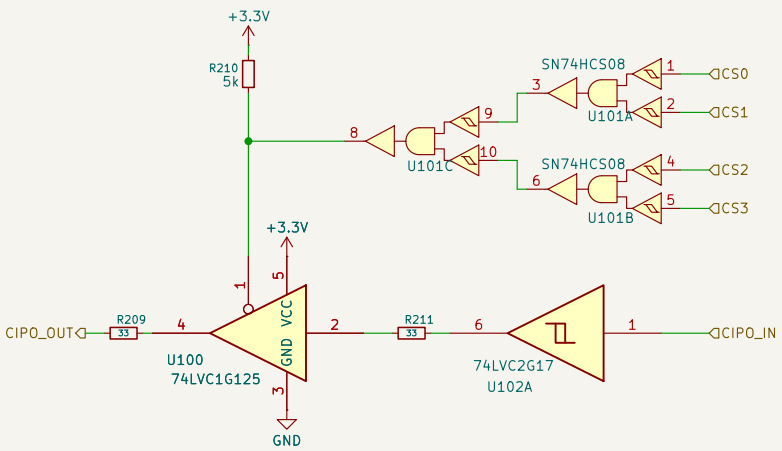


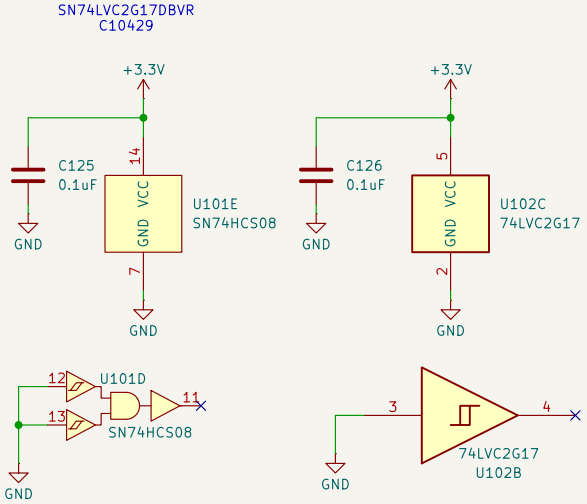







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|---|---|
| **Reiser Lab @ Janelia \| HHMI** | |
| Sheet: /Fan Out/CIPO B0 Enable for P4/ | |
| File: cipo_enable.kicad_sch | |
| ***Title: G6 Teensy 10–14 Arena*** | |
| Size: USLetter \| Date: 2026-10-25 | **Rev: v1.0** |
| KiCad E.D.A. 10.0.7 | Id: 57/67 |

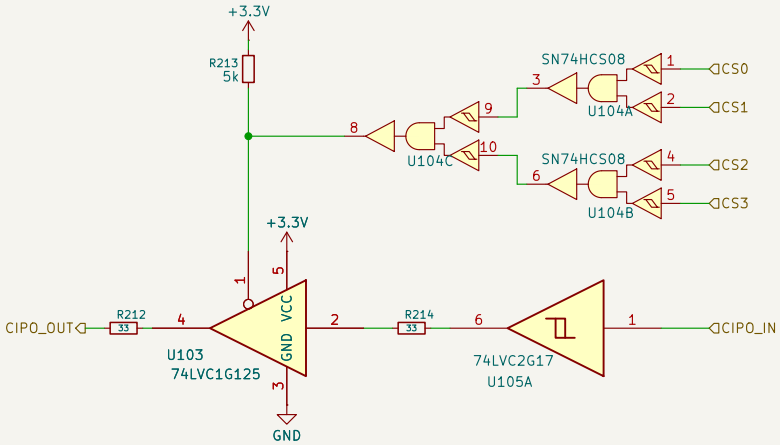






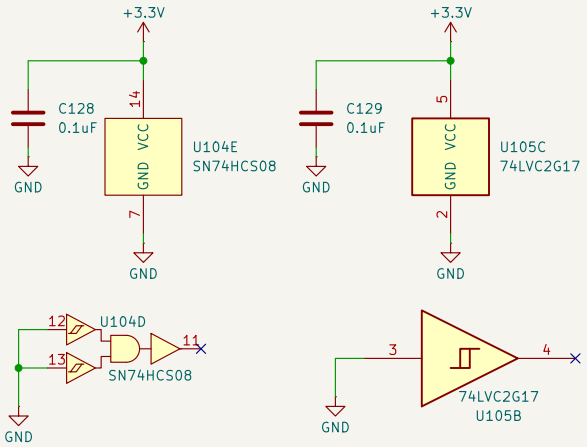
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|---|---|
| **Reiser Lab @ Janelia \| HHMI** | |
| Sheet: /Fan Out/CIPO B0 Enable for P5/ | |
| File: cipo_enable.kicad_sch | |
| ***Title: G6 Teensy 10–14 Arena*** | |
| Size: USLetter \| Date: 2026-10-25 | **Rev: v1.0** |
| KiCad E.D.A. 10.0.7 | Id: 58/67 |



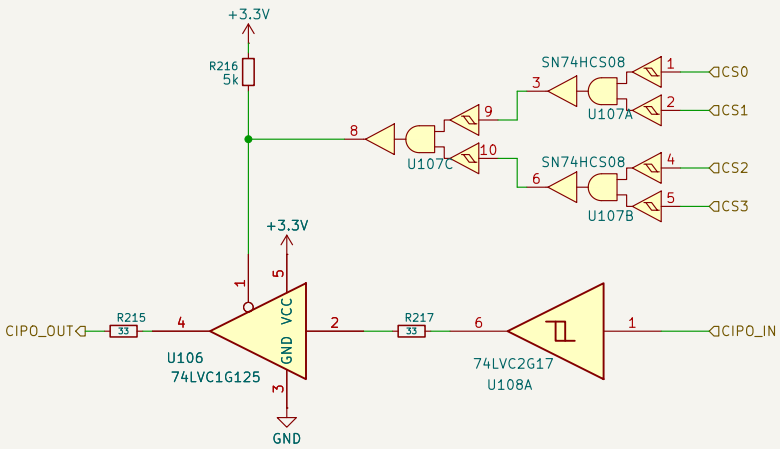


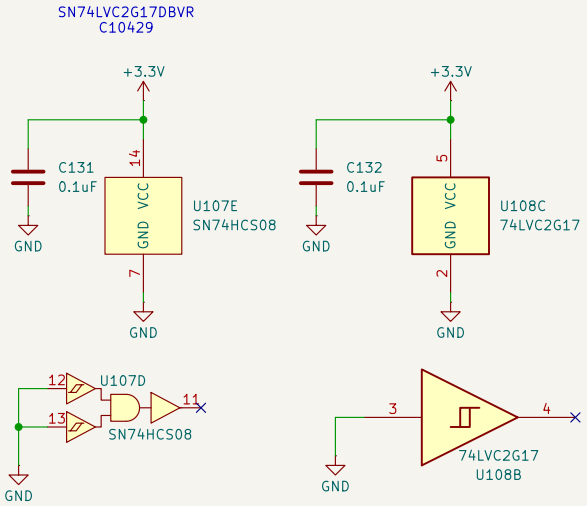


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C10429




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| **Reiser Lab @ Janelia \| HHMI** | |
| Sheet: /Fan Out/CIPO B1 Enable for P7/ | |
| File: cipo_enable.kicad_sch | |
| ***Title: G6 Teensy 10–14 Arena*** | |
| Size: USLetter \| Date: 2026-10-25 | **Rev: v1.0** |
| KiCad E.D.A. 10.0.7 | Id: 59/67 |







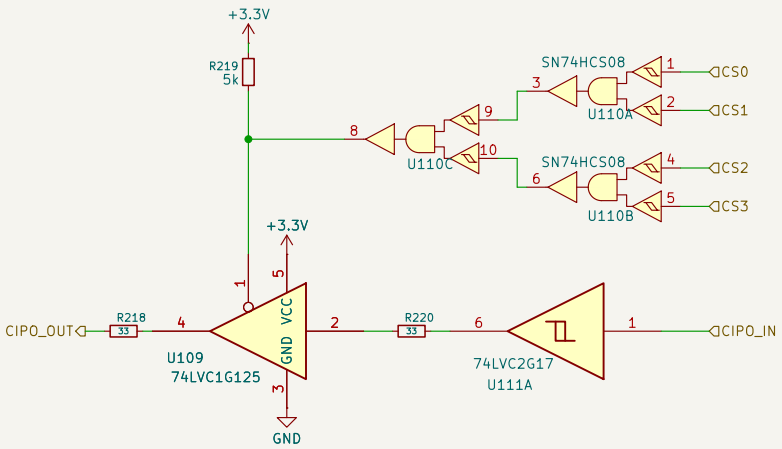
Reiser Lab @ Janelia | HHMI

Sheet: /Fan Out/CIPO B1 Enable for P8/

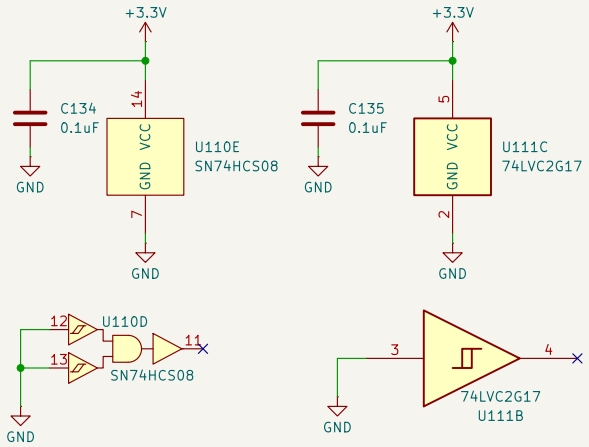
File: cipo_enable.kicad_sch

**Title: G6 Teensy 10-14 Arena**

| Size: USLetter | Date: 2026-10-25 | **Rev: v1.0** |
|---|---|---|
| KiCad E.D.A. 10.0.7 | | Id: 60/67 |




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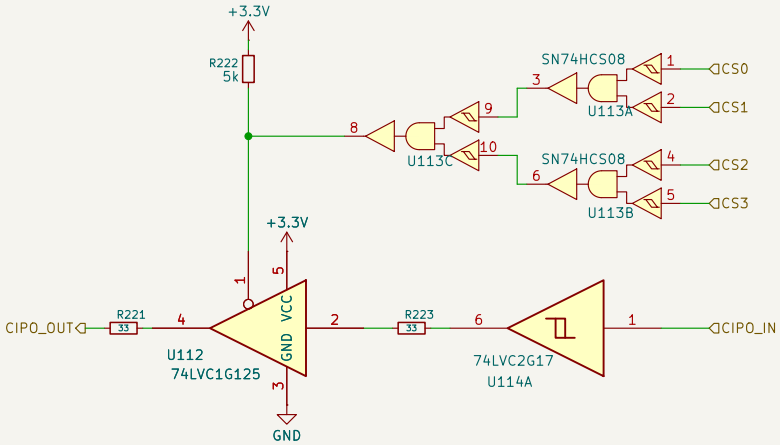


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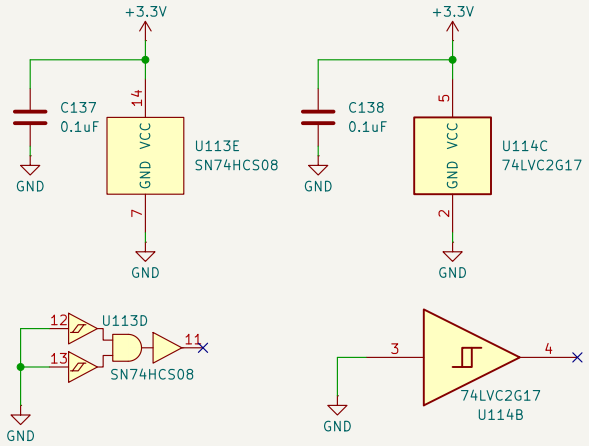
Sheet: /Fan Out/CIPO B1 Enable for P9/
File: cipo_enable.kicad_sch

***Title: G6 Teensy 10–14 Arena***

| Size: USLetter | Date: 2026-10-25 | **Rev: v1.0** |
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| KiCad E.D.A. 10.0.7 | | Id: 61/67 |




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C10429





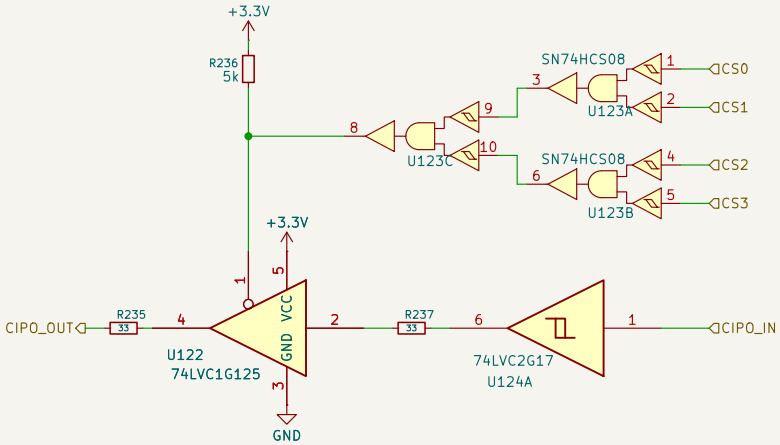


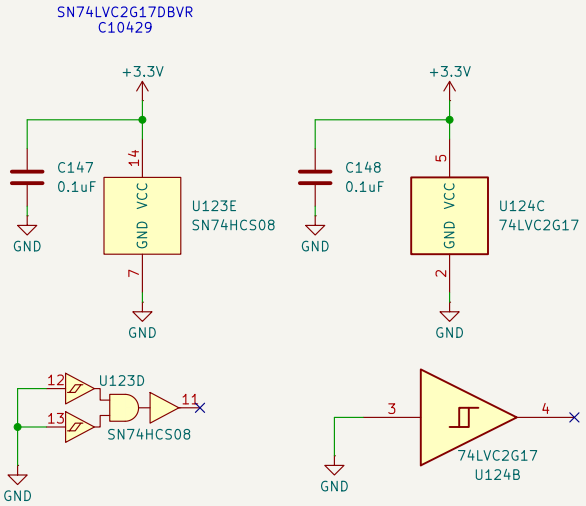
**Reiser Lab @ Janelia | HHMI**

Sheet: /Fan Out/CIPO B1 Enable for P10/
File: cipo_enable.kicad_sch

***Title: G6 Teensy 10–14 Arena***

| Size: USLetter | Date: 2026-10-25 | **Rev: v1.0** |
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| KiCad E.D.A. 10.0.7 | | Id: 62/67 |







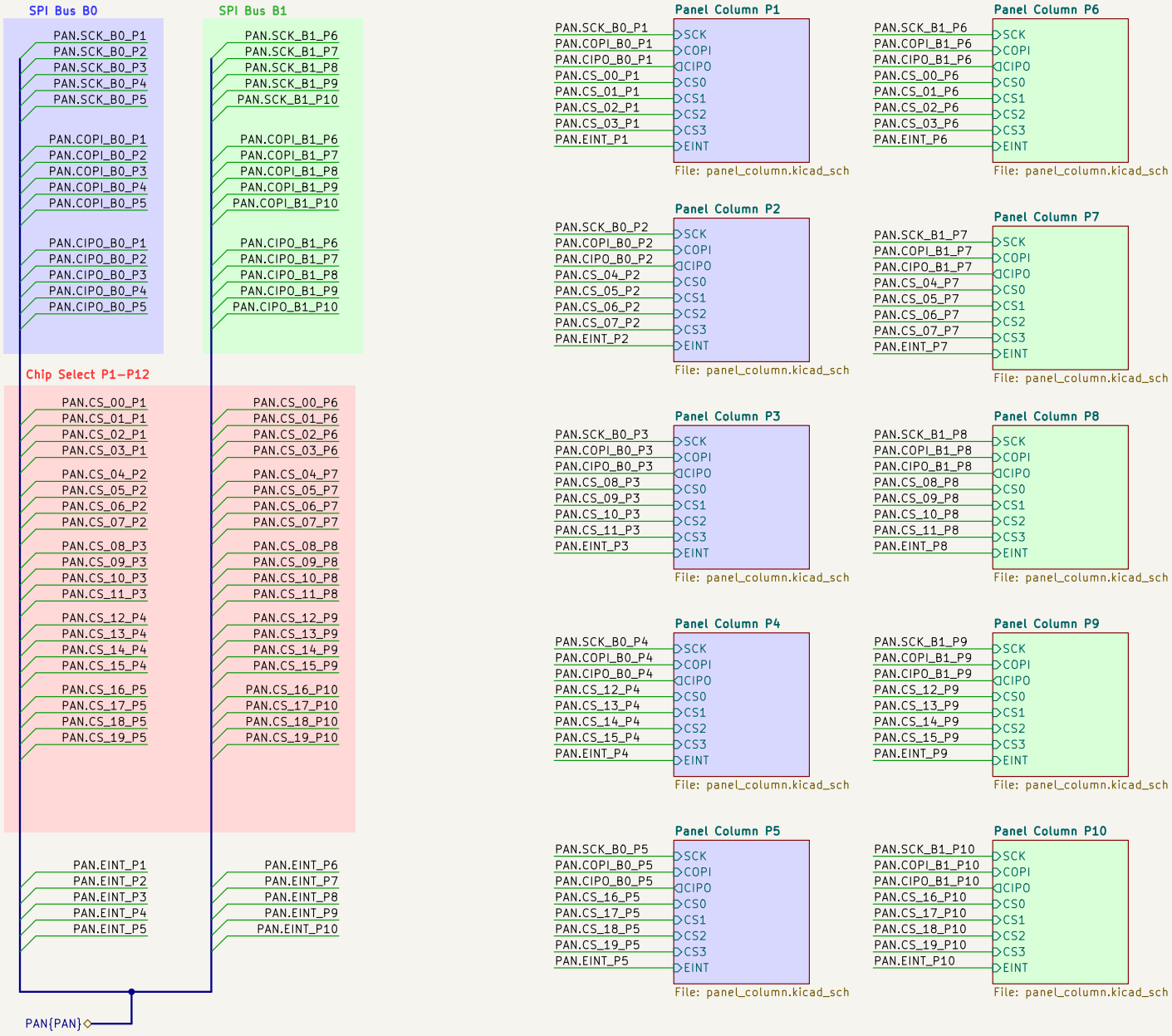
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Sheet: /Fan Out/CIPO B1 Enable for P6/

File: cipo_enable.kicad_sch

**Title: G6 Teensy 10-14 Arena**

| Size: USLetter | Date: 2026-10-25 | **Rev: v1.0** |
|---|---|---|
| KiCad E.D.A. 10.0.7 | | Id: 68/67 |



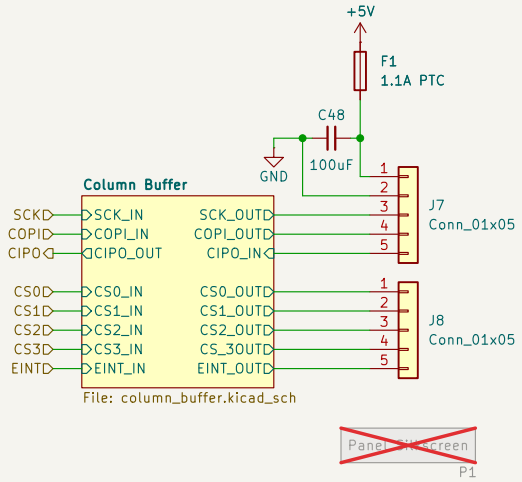

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Sheet: /Panels/

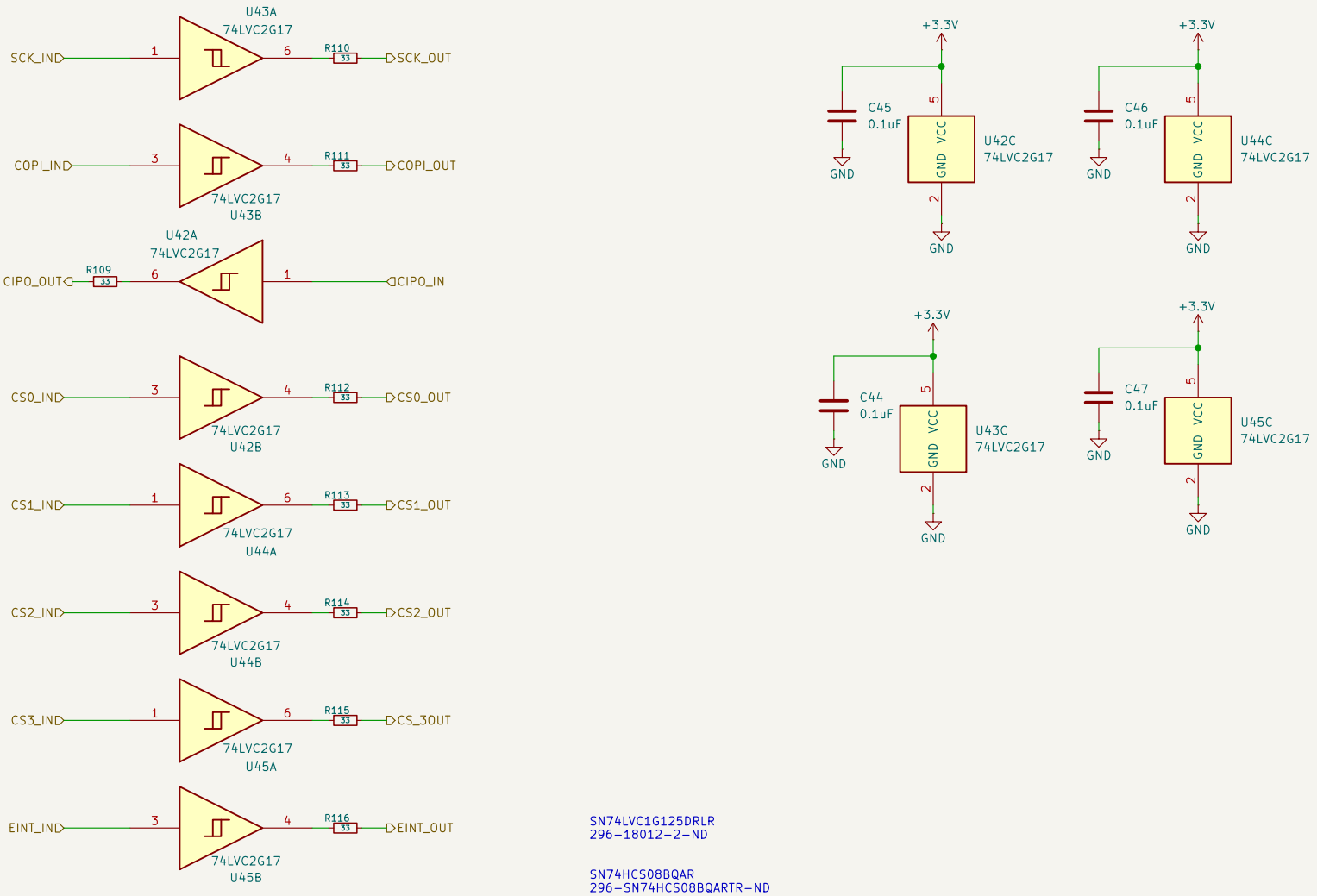
File: panels.kicad_sch

**Title: G6 Teensy 10–14 Arena**

| Size: USLetter | Date: 2026-10-25 | Rev: v1.0 |
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| KiCad E.D.A. 10.0.7 | | Id: 30/67 |




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| **Reiser Lab @ Janelia \| HHMI** | |
| Sheet: /Panels/Panel Column P1/ | |
| File: panel_column.kicad_sch | |
| ***Title: G6 Teensy 10–14 Arena*** | |
| Size: USLetter \| Date: 2026-10-25 | **Rev: v1.0** |
| KiCad E.D.A. 10.0.7 | Id: 32/67 |




SN74LVC1G125DRLR
296-18012-2-ND

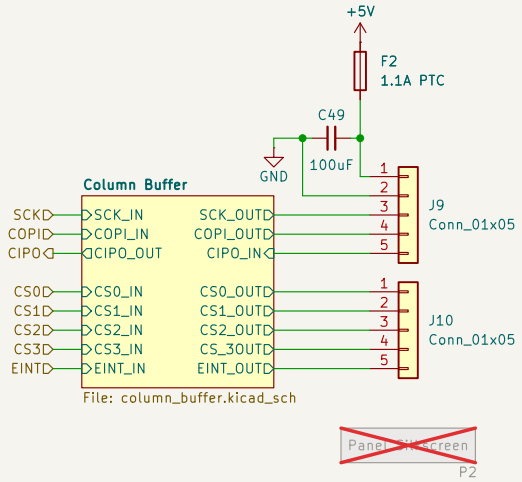
SN74HCS08BQAR
296-SN74HCS08BQARTR-ND

**Reiser Lab @ Janelia | HHMI**

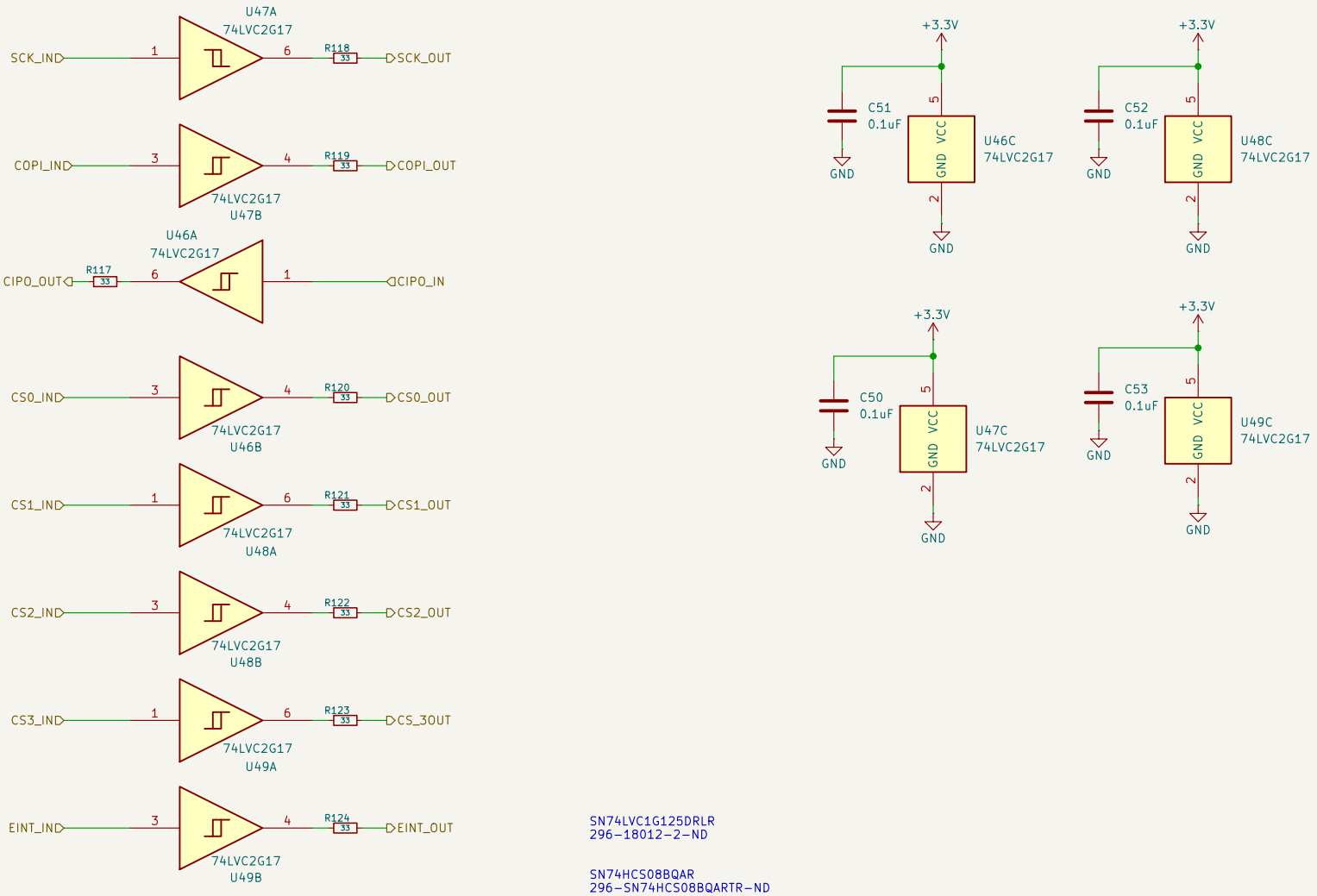
Sheet: /Panels/Panel Column P1/Column Buffer/
File: column_buffer.kicad_sch

***Title: G6 Teensy 10-14 Arena***

| Size: USLetter | Date: 2026-10-25 | **Rev: v1.0** |
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| KiCad E.D.A. 10.0.7 | | Id: 31/67 |




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| Reiser Lab @ Janelia \| HHMI | |
| Sheet: /Panels/Panel Column P2/ | |
| File: panel_column.kicad_sch | |
| **Title: G6 Teensy 10–14 Arena** | |
| Size: USLetter \| Date: 2026-10-25 | **Rev: v1.0** |
| KiCad E.D.A. 10.0.7 | Id: 33/67 |




SN74LVC1G125DRLR
296-18012-2-ND

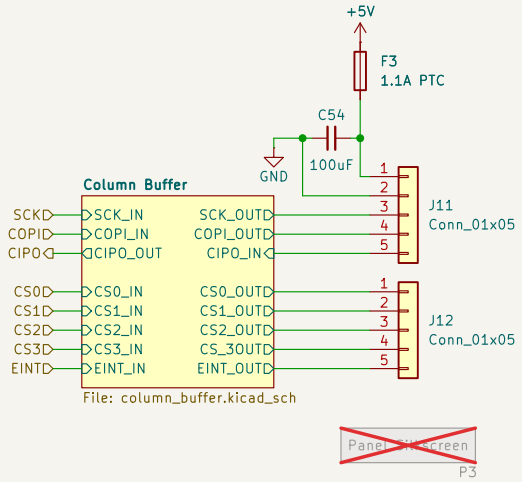
SN74HCS08BQAR
296-SN74HCS08BQARTR-ND

**Reiser Lab @ Janelia | HHMI**

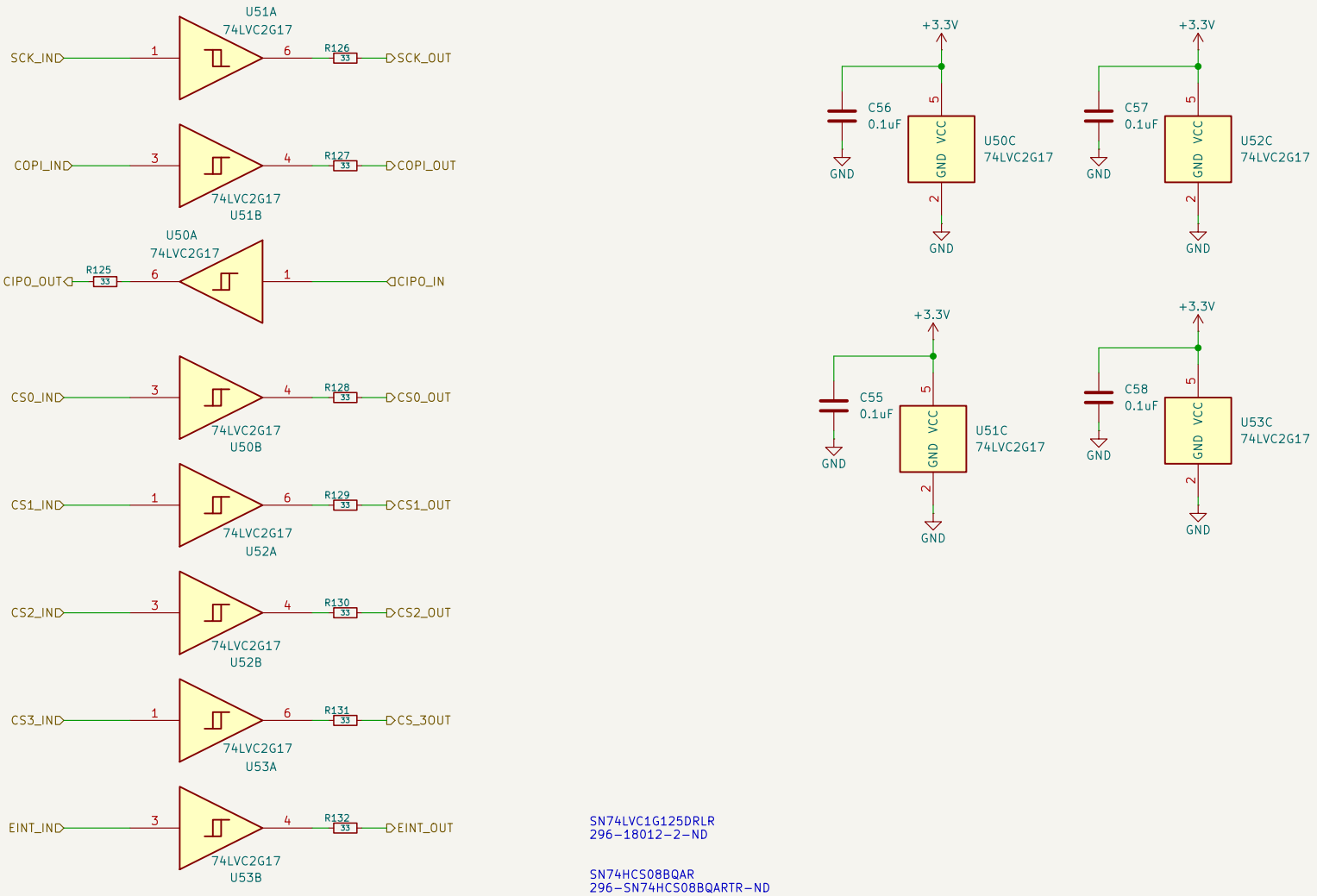
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File: column_buffer.kicad_sch

***Title: G6 Teensy 10-14 Arena***

| Size: USLetter | Date: 2026-10-25 | **Rev: v1.0** |
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| KiCad E.D.A. 10.0.7 | | Id: 34/67 |




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| Reiser Lab @ Janelia \| HHMI | |
| Sheet: /Panels/Panel Column P3/ | |
| File: panel_column.kicad_sch | |
| **Title: G6 Teensy 10–14 Arena** | |
| Size: USLetter \| Date: 2026-10-25 | **Rev: v1.0** |
| KiCad E.D.A. 10.0.7 | Id: 35/67 |







SN74LVC1G125DRLR
296-18012-2-ND

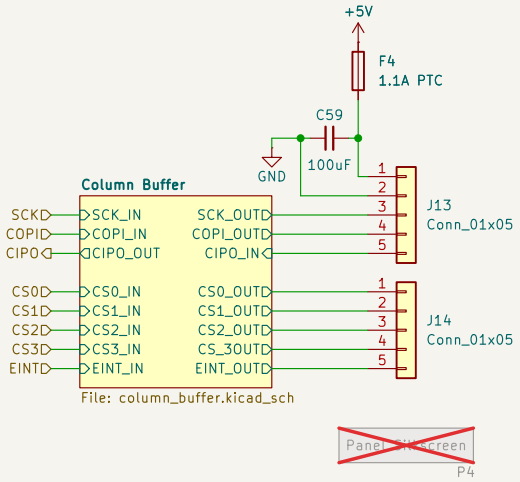
SN74HCS08BQAR
296-SN74HCS08BQARTR-ND

**Reiser Lab @ Janelia | HHMI**

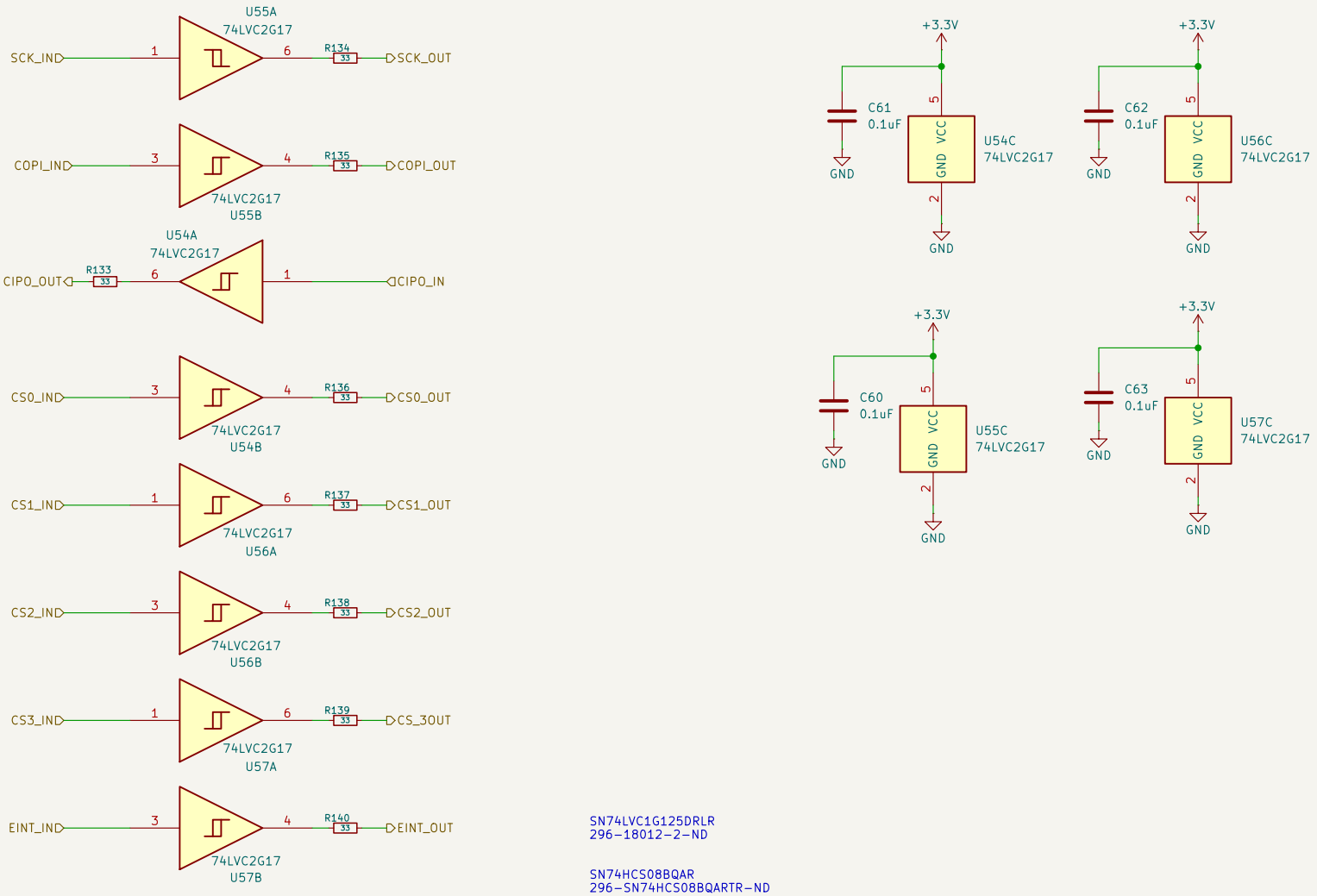
Sheet: /Panels/Panel Column P3/Column Buffer/
File: column_buffer.kicad_sch

***Title: G6 Teensy 10-14 Arena***

| Size: USLetter | Date: 2026-10-25 | **Rev: v1.0** |
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| KiCad E.D.A. 10.0.7 | | Id: 36/67 |




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| **Reiser Lab @ Janelia \| HHMI** | |
| Sheet: /Panels/Panel Column P4/ | |
| File: panel_column.kicad_sch | |
| ***Title: G6 Teensy 10–14 Arena*** | |
| Size: USLetter \| Date: 2026-10-25 | **Rev: v1.0** |
| KiCad E.D.A. 10.0.7 | Id: 37/67 |







SN74LVC1G125DRLR
296-18012-2-ND

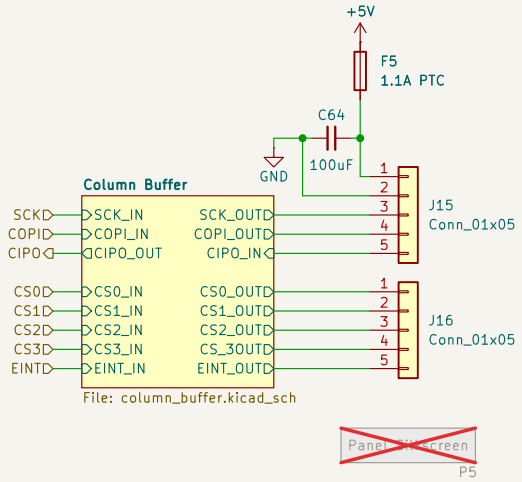
SN74HCS08BQAR
296-SN74HCS08BQARTR-ND

**Reiser Lab @ Janelia | HHMI**

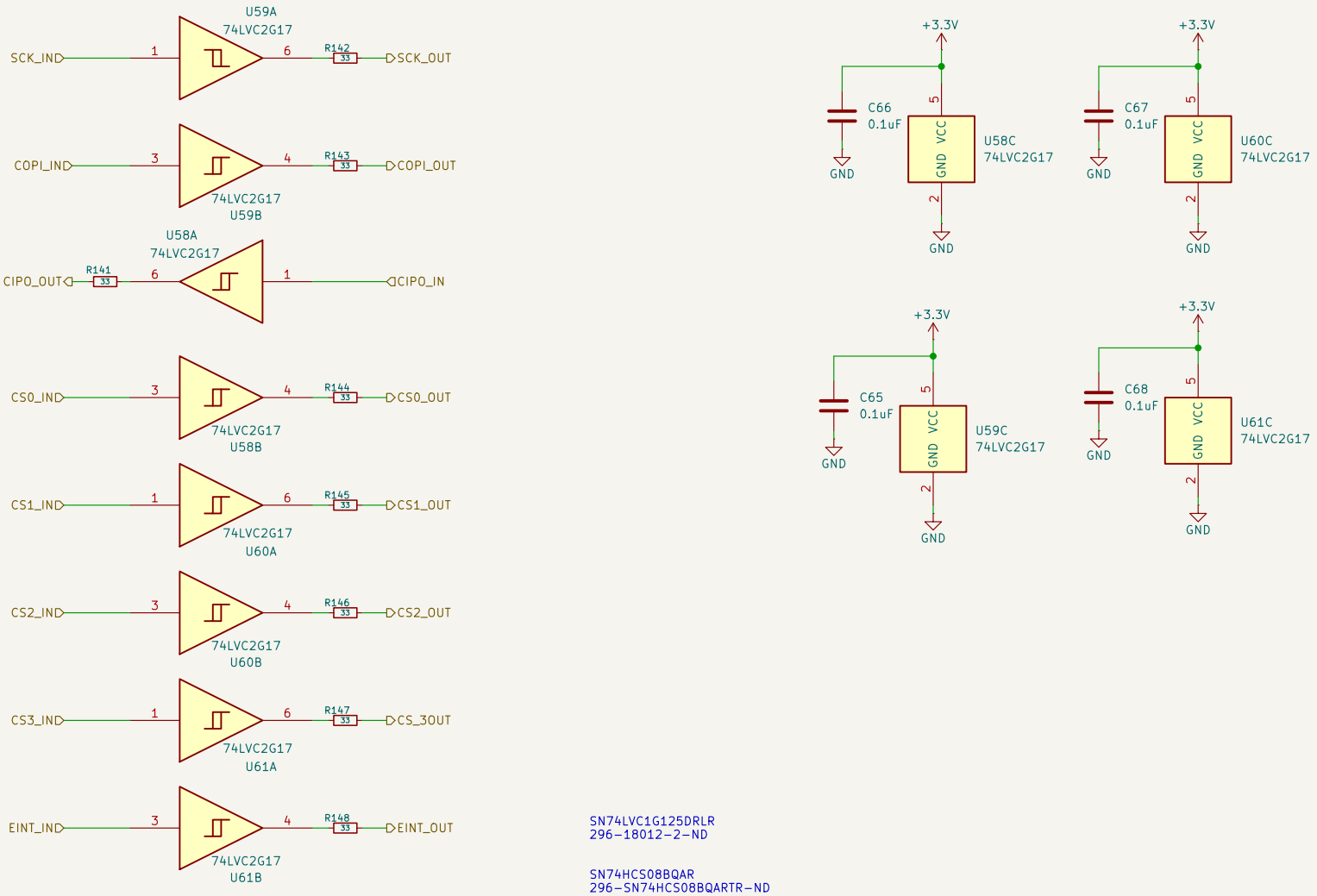
Sheet: /Panels/Panel Column P4/Column Buffer/
File: column_buffer.kicad_sch

***Title: G6 Teensy 10-14 Arena***

| Size: USLetter | Date: 2026-10-25 | **Rev: v1.0** |
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| KiCad E.D.A. 10.0.7 | | Id: 38/67 |




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| Reiser Lab @ Janelia \| HHMI | |
| Sheet: /Panels/Panel Column P5/ | |
| File: panel_column.kicad_sch | |
| **Title: G6 Teensy 10–14 Arena** | |
| Size: USLetter \| Date: 2026-10-25 | **Rev: v1.0** |
| KiCad E.D.A. 10.0.7 | Id: 39/67 |




SN74LVC1G125DRLR
296-18012-2-ND

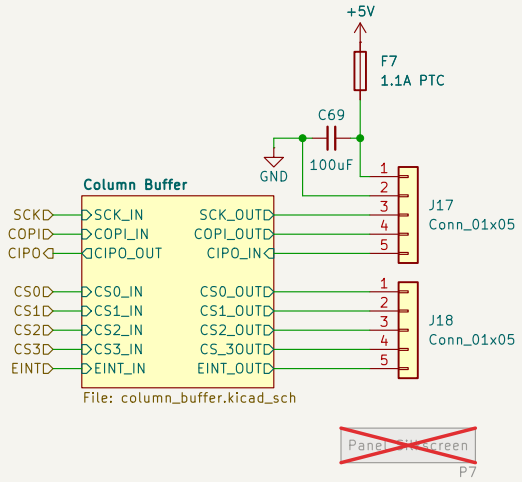
SN74HCS08BQAR
296-SN74HCS08BQARTR-ND

**Reiser Lab @ Janelia | HHMI**

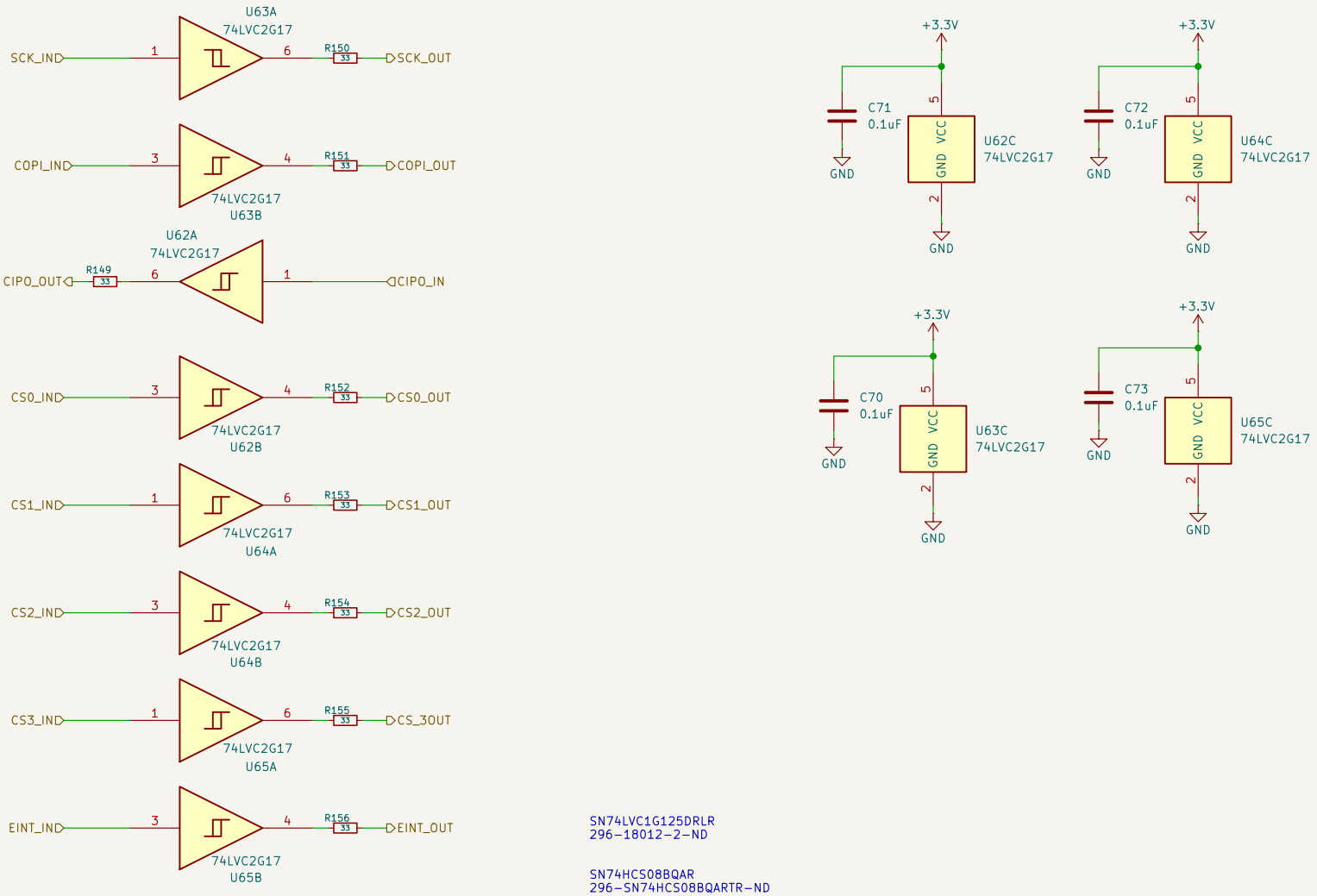
Sheet: /Panels/Panel Column P5/Column Buffer/
File: column_buffer.kicad_sch

***Title: G6 Teensy 10-14 Arena***

| Size: USLetter | Date: 2026-10-25 | **Rev: v1.0** |
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| KiCad E.D.A. 10.0.7 | | Id: 40/67 |




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| Reiser Lab @ Janelia \| HHMI | |
| Sheet: /Panels/Panel Column P7/ | |
| File: panel_column.kicad_sch | |
| **Title: G6 Teensy 10–14 Arena** | |
| Size: USLetter \| Date: 2026-10-25 | **Rev: v1.0** |
| KiCad E.D.A. 10.0.7 | Id: 41/67 |




SN74LVC1G125DRLR
296-18012-2-ND

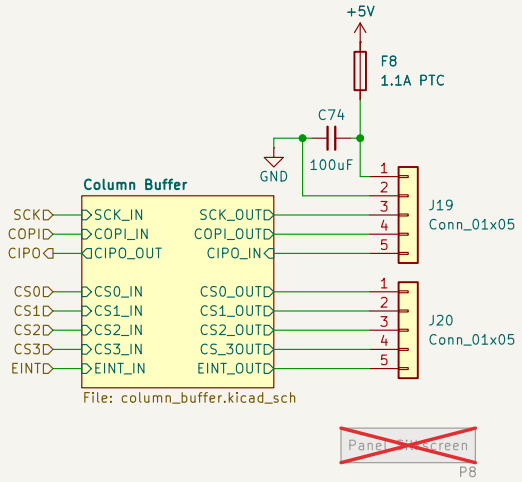
SN74HCS08BQAR
296-SN74HCS08BQARTR-ND

**Reiser Lab @ Janelia | HHMI**

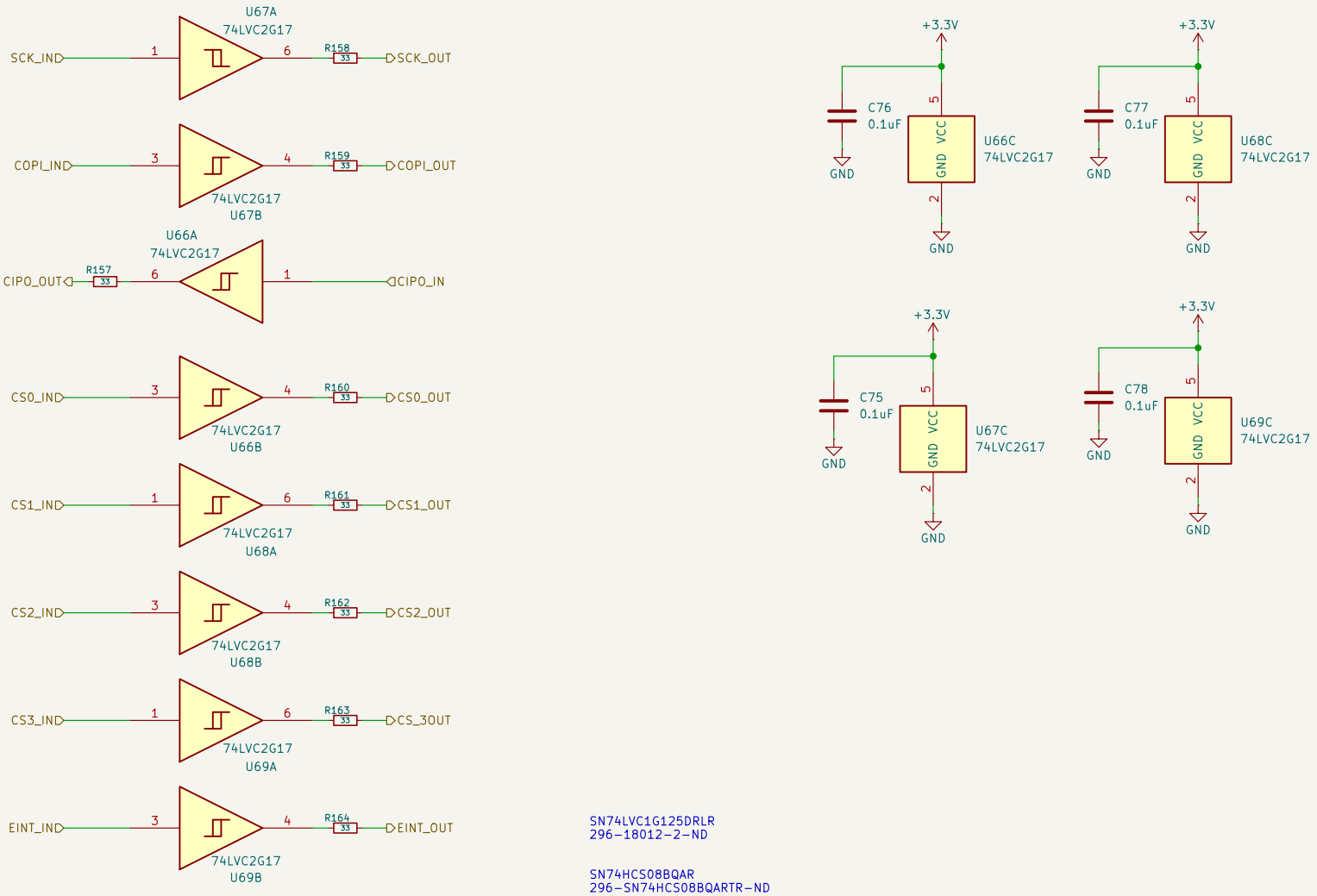
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File: column_buffer.kicad_sch

***Title: G6 Teensy 10-14 Arena***

| Size: USLetter | Date: 2026-10-25 | **Rev: v1.0** |
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| KiCad E.D.A. 10.0.7 | | Id: 42/67 |




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| Reiser Lab @ Janelia \| HHMI | |
| Sheet: /Panels/Panel Column P8/ | |
| File: panel_column.kicad_sch | |
| **Title: G6 Teensy 10–14 Arena** | |
| Size: USLetter \| Date: 2026-10-25 | **Rev: v1.0** |
| KiCad E.D.A. 10.0.7 | Id: 43/67 |




SN74LVC1G125DRLR
296-18012-2-ND

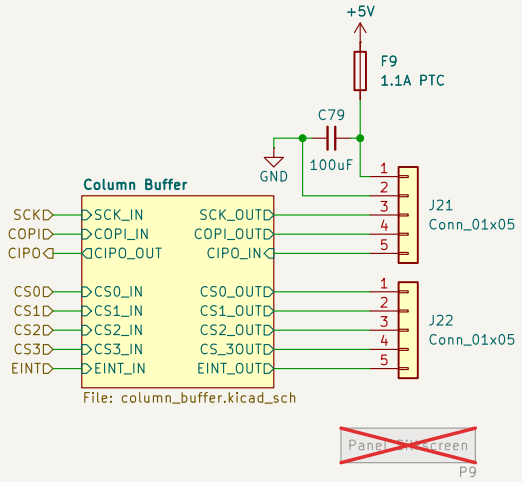
SN74HCS08BQAR
296-SN74HCS08BQARTR-ND

**Reiser Lab @ Janelia | HHMI**

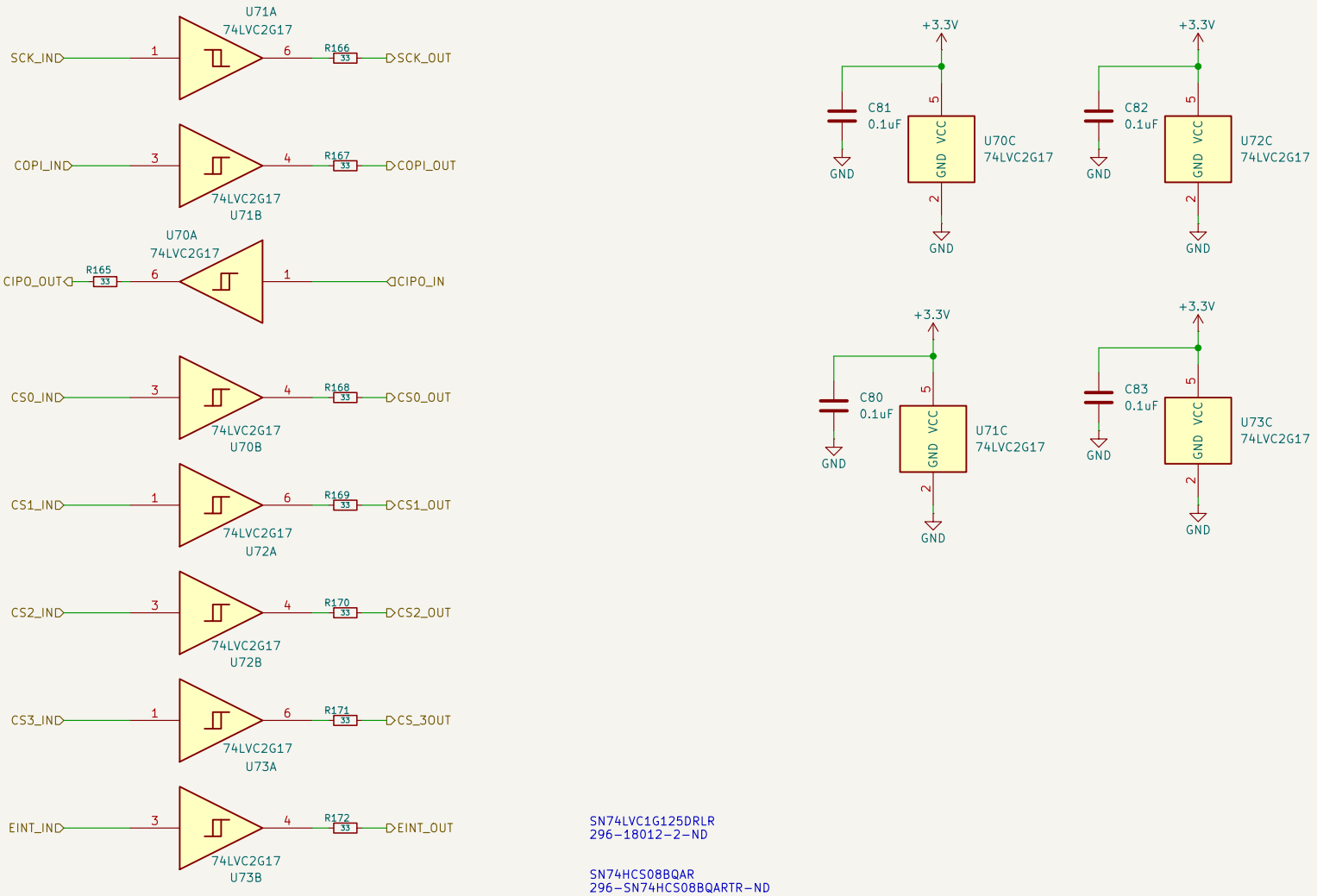
Sheet: /Panels/Panel Column P8/Column Buffer/
File: column_buffer.kicad_sch

***Title: G6 Teensy 10-14 Arena***

| Size: USLetter | Date: 2026-10-25 | **Rev: v1.0** |
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| KiCad E.D.A. 10.0.7 | | Id: 44/67 |




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| Reiser Lab @ Janelia \| HHMI | |
| Sheet: /Panels/Panel Column P9/ | |
| File: panel_column.kicad_sch | |
| **Title: G6 Teensy 10–14 Arena** | |
| Size: USLetter \| Date: 2026-10-25 | **Rev: v1.0** |
| KiCad E.D.A. 10.0.7 | Id: 45/67 |




SN74LVC1G125DRLR
296-18012-2-ND

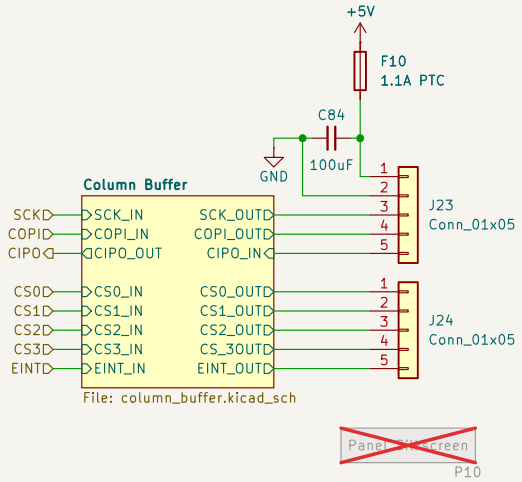
SN74HCS08BQAR
296-SN74HCS08BQARTR-ND

**Reiser Lab @ Janelia | HHMI**

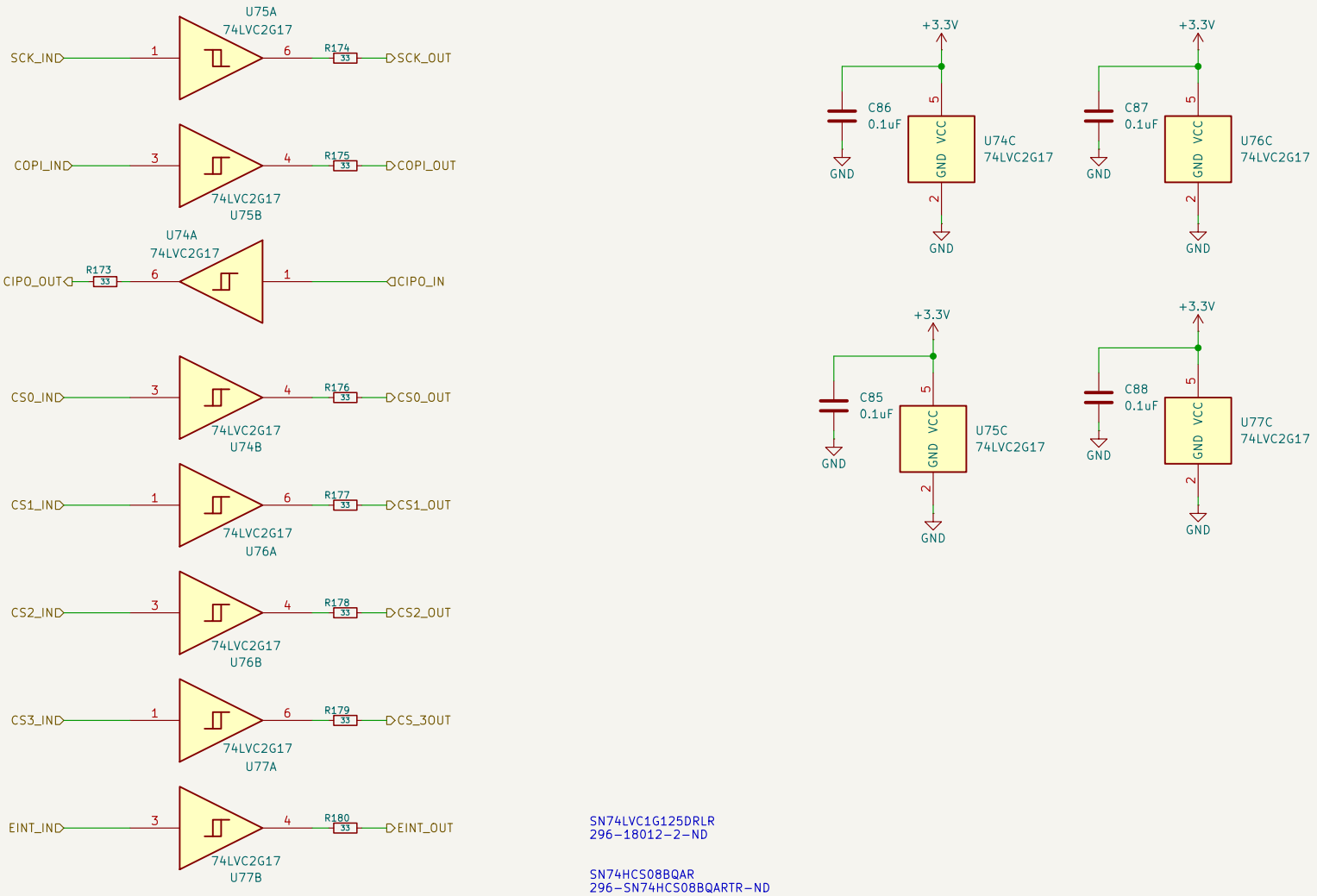
Sheet: /Panels/Panel Column P9/Column Buffer/
File: column_buffer.kicad_sch

***Title: G6 Teensy 10-14 Arena***

| Size: USLetter | Date: 2026-10-25 | **Rev: v1.0** |
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| KiCad E.D.A. 10.0.7 | | Id: 46/67 |




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| Reiser Lab @ Janelia \| HHMI | |
| Sheet: /Panels/Panel Column P10/ | |
| File: panel_column.kicad_sch | |
| **Title: G6 Teensy 10–14 Arena** | |
| Size: USLetter \| Date: 2026-10-25 | **Rev: v1.0** |
| KiCad E.D.A. 10.0.7 | Id: 47/67 |







SN74LVC1G125DRLR
296-18012-2-ND

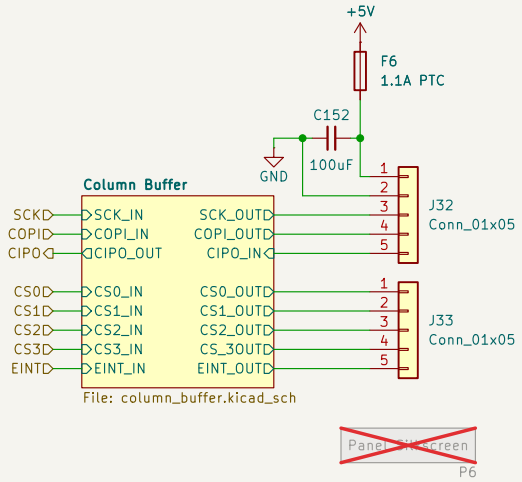
SN74HCS08BQAR
296-SN74HCS08BQARTR-ND

**Reiser Lab @ Janelia | HHMI**

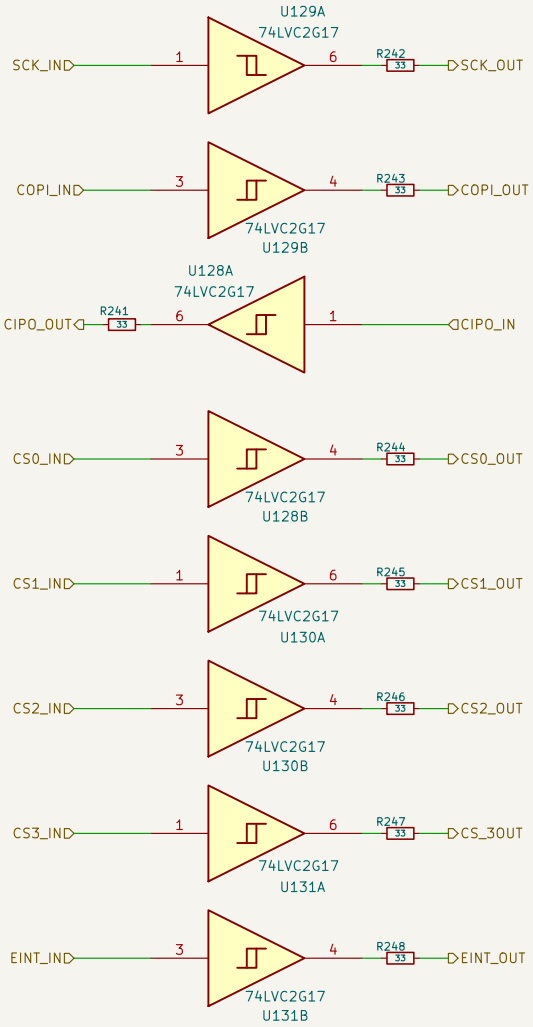
Sheet: /Panels/Panel Column P10/Column Buffer/
File: column_buffer.kicad_sch

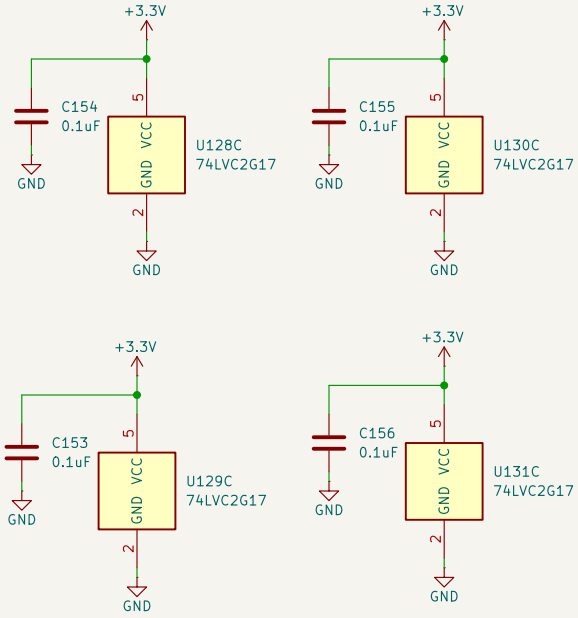
***Title: G6 Teensy 10-14 Arena***

| Size: USLetter | Date: 2026-10-25 | **Rev: v1.0** |
|---|---|---|
| KiCad E.D.A. 10.0.7 | | Id: 48/67 |




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| Reiser Lab @ Janelia \| HHMI | |
| Sheet: /Panels/Panel Column P6/ | |
| File: panel_column.kicad_sch | |
| **Title: G6 Teensy 10–14 Arena** | |
| Size: USLetter \| Date: 2026-10-25 | **Rev: v1.0** |
| KiCad E.D.A. 10.0.7 | Id: 70/67 |




SN74LVC1G125DRLR
296-18012-2-ND

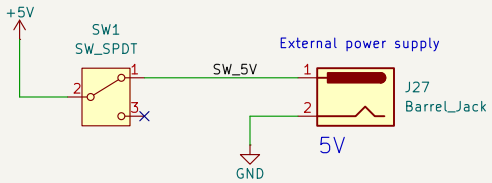
SN74HCS08BQAR
296-SN74HCS08BQARTR-ND

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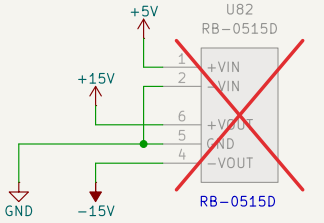
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***Title: G6 Teensy 10-14 Arena***

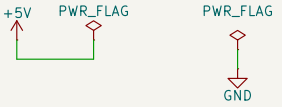
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| KiCad E.D.A. 10.0.7 | | Id: 71/67 |

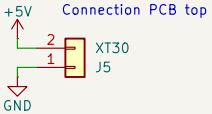


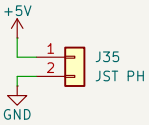

Change DC Jack to
PN DC-053D-20A Korean Hroparts Elec
LCSC C707274




Power rails for
bipolar analog inputs



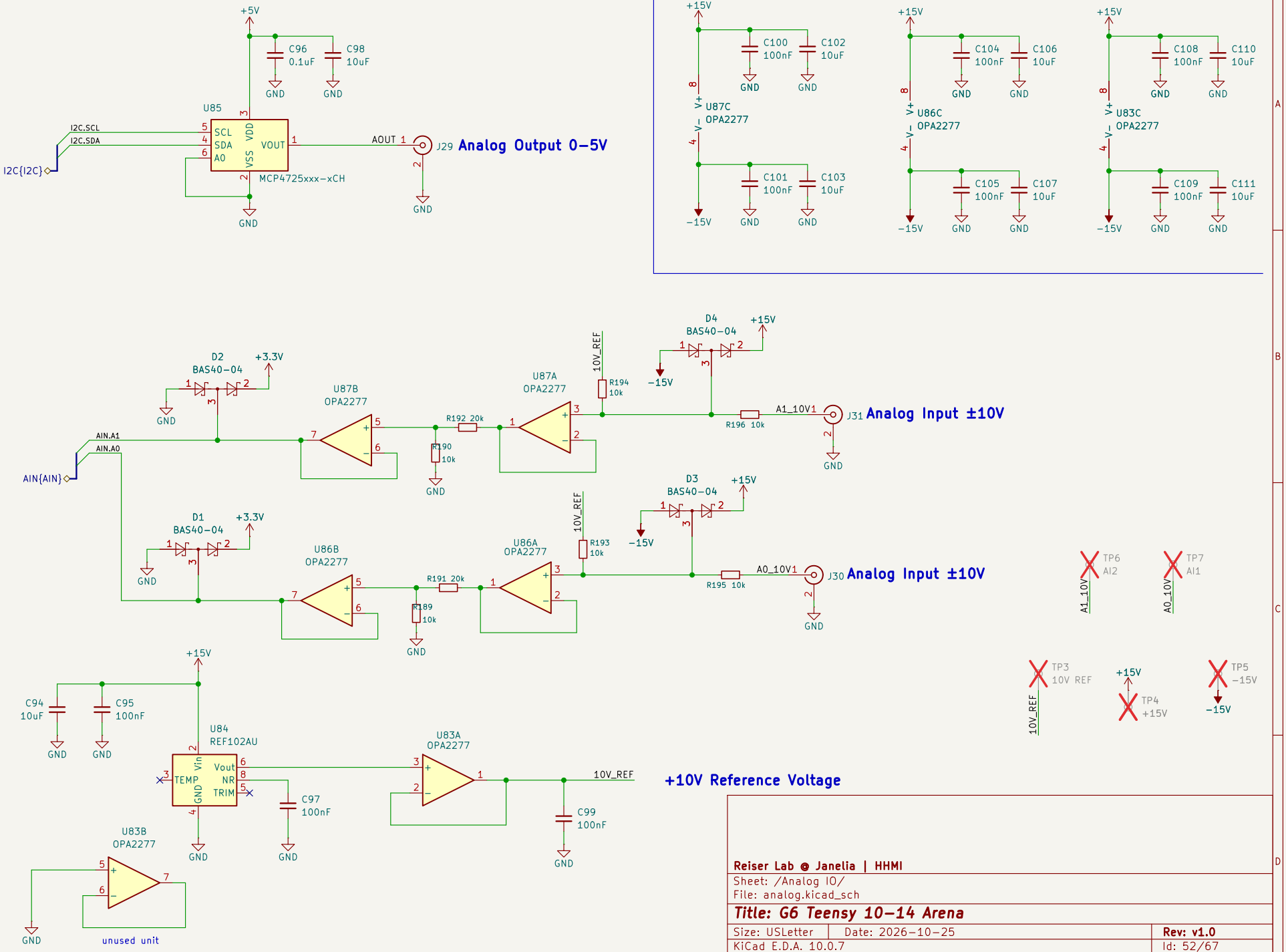




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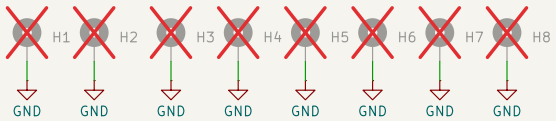
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***Title: G6 Teensy 10-14 Arena***

| | | |
|---|---|---|
| Size: USLetter | Date: 2026-10-25 | **Rev: v1.0** |
| KiCad E.D.A. 10.0.7 | | Id: 51/67 |




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|---|---|
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| Sheet: /Analog IO/ | |
| File: analog.kicad_sch | |
| **Title: G6 Teensy 10–14 Arena** | |
| Size: USLetter \| Date: 2026-10-25 | **Rev: v1.0** |
| KiCad E.D.A. 10.0.7 | Id: 52/67 |










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Sheet: /Mount Holes/
File: holes.kicad_sch

***Title: G6 Teensy 10–14 Arena***

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|---|---|---|
| Size: USLetter | Date: 2026-10-25 | **Rev: v1.0** |
| KiCad E.D.A. 10.0.7 | | Id: 53/67 |